

Viability of decentralized decision-making within Curaçao's institutions



DAOstack



CARIBBEAN
BLOCKCHAIN NETWORK

Colophon

This report is the result of a research initiative initialized by the **Cairbbean Blockchain Network (CBN)** and facilitated by **DAOstack's**, **Alchemy** and **Genesis DAO** -the first pilot DAO created through Arc - (a smart contract framework developed by **DAOstack**).

Two of the contributors are active members of the Caribbean Blockchain Network (CBN) and the rest of the contributors are individuals that supported the research through a bounty initiative proposed by the members of CBN in the **Genesis DAO**.

Contributor	Contributions
Ruben Russel	Introduction, Theoretical Background (all sections), Results, Conclusions, Executive Summary
Leroy Gilbert	Environmental Sphere
Oneisey Rosini	Socio-political Sphere
Ayrton Sambo	Socio-economic Sphere
Fernando Trouw	Introduction, Theoretical Background (all sections), Results, Conclusions, Executive Summary and report editing

Publisher: www.caribbeanblockchain.net

Publish date: 30th of October 2019

Executive Summary

Curaçao small island nation in the Caribbean, just like any other nation in the world is home to many social problems that stem from the way economic organizing is realized i.e. the use of centralized decision making in institutions that are tasked with the realization of the welfare of all of the inhabitants of the island. These problems range from inequality, to poverty to crime. It has been postulated that economic organizing through decentralized autonomous organizing could prove to be a potential solution to these problems.

Therefore, a research project was set up with the goal to identify the major shortcomings (negative socio-economic problems) that are related to centralized decision-making structures (public and private institutions) in Curaçao in the past ten years. Thereupon, a conceptual framework for a DAO that could potentially result in the mitigation of the identified shortcomings was designed.

The findings suggest that there are shortcomings in all of the spheres that were analyzed, which include: transparency, accountability, cooperation and corruption. Furthermore, these shortcomings translate into the following social issues (symptoms) such as, social misrepresentation, nepotism and populism.

From the analysis of the shortcomings and the possible solutions proposed by DAOs, a conceptual design was created. The conceptual design was based on three layers, the blockchain layer, the application layer and the DAO layer. Jointly, by design, the concept showed potential solutions to the shortcomings related to transparency, accountability, corruption and the problems that are a consequence of those shortcomings e.g. social misrepresentation, nepotism and populism.

Table of Contents

Colophon	2
Executive Summary	3
1 Introduction.....	1
1.1 Goal	2
1.2 Boundaries	2
1.3 Reading Guide	2
2 Theoretical Background	4
2.1 Traditional organizing (Hierarchical & centralized decision making)	4
2.2 Decentralized organizations	5
2.2.1 Reasons for decentralized organizing	5
2.3 Decentralized autonomous organizations	6
2.3.1 History of Decentralized Autonomous Organizations	6
2.4 Benefits of the DAO	9
2.5 The drawbacks of a DAO	10
2.5.1 Class and role of the state	10
2.5.2 Concentration and centralization of capital	12
2.5.3 Automation.....	12
2.6 Smart contract frameworks for DAO's.....	13
2.6.1 Colony	13
2.6.2 DAOstack's Arc.....	16
2.6.3 Aragon	17
2.7 Consequences of economic organizing on Curaçao	20
2.7.1 Socio Political sphere	20
2.7.2 Socio-economic sphere	32
2.7.3 Environmental sphere	34
3 Methodology	37
3.1 Theory analysis	37

3.2	Theoretical DAO concept	37
4	Results	39
4.1	Theory analysis	39
4.1.1	Socio-political sphere.....	40
4.1.2	Socio-economic sphere	41
4.1.3	Environmental Sphere	42
4.1.4	Connecting the dots.....	42
4.2	Theoretical DAO concept	43
4.2.1	Distributed trust network (blockchain)	44
4.2.2	Application Layer (smart contract layer)	45
4.2.3	Decentralized Autonomous Organization (DAO).....	45
	Discussion.....	47
	Conclusion	48
	References.....	49



1 Introduction

Curaçao is an island nation that is located in the north-east of Latin America. It is a small nation which according to the world population review has a population of around 160 000 people (Worldpopulationreview, 2019). Even though Curaçao is a small island with a relatively small population, it is, just like many other islands in the Caribbean, experiencing many socio-economic problems. For instance, according to the most recent census, around 25 % of the households in Curaçao live under the poverty line (Centraal Bureau voor de Statistiek, 2008). Furthermore, Curaçao is considered to have a very high unemployment rate, highest amongst all of the Caribbean islands (14.5 %) (Matroos-Lasten, 2018). which has exacerbated crime on the island. According to the Overseas Security Advisory Council (OSAC), between the years 2014-2015 private law enforcement officials estimate that violent crime has increased, especially with the rising youth unemployment (estimated at over 35%). Moreover, one out of three people (25% of men, 38% of women) in Curaçao have experienced some form of domestic violence at some point in their adult lives.

We think the above stated issues are for a large part associated to how the human basic needs are organized on the island: in essence, the way the economy is organized. This is why we have chosen to look into alternative and new experimental organizational models that could offer more positive socio-economic outcomes. The reasons why human individuals or groups organize themselves is to achieve certain goals. Whether it is for having clean and fresh water for a group of humans that organize themselves into a country or the energy that is needed to fuel production of food, shelter or transportation. Organizing is highly necessary to reach any form of human basic needs. It is at this level that the socio-economic problems that are currently being experienced in Curaçao are not just leadership problems but organizational problems (Bril & Bessems, 2016).

Herewith, when looking at the fundamental underlying principles with which we organize our society, it seems that most organizational models are still based on models dating back to the beginning of the twentieth century. They are especially suitable for physical production in large numbers and a relatively quiet local market. Traditionally, we organize supply and demand through organizations where decisions are made by: management, shareholders and/or a board of directors in a centralized organizational structure (ex. private companies, traditional markets or government). Yet, these forms of organizations bring with them potential organizational problems e.g. rapidly increasing transactional and management costs and declining productivity growth, concentration of power, unequal distribution of: work, income and wealth, destructive populism resource depletion and global warming (Hutt, 2016).

To tackle these well-known fundamental socio-economic problems, one could explore a new way of organizing: *“organizing by design”* this means that we primarily do not focus on adjusting the existing (often referred to as 'Brownfield' organizing), but on fundamental innovation and change (often referred to as 'Greenfield' organizing). A powerful method to redesign the organizational problems we are facing today is *“Blockchain organizing”*. This method was introduced in the book, *Blockchain Organiseren*, which aims to tackle the root cause of our current socio-economic problems (organizing supply and demand efficiently). Blockchain organizing utilizes Blockchain technology protocols and the applications that are built on these protocols to facilitate a new way to design efficient and frictionless alternative organizational models such as Decentralized Autonomous



Organizations (DAOs). This new method of organizing supply and demand could fundamentally tackle these common traditional organizational deficiencies.

1.1 Goal

The main goal of this research project is to identify major negative socio-economic problems that are related to centralized decision-making structures (public and private institutions) in Curaçao in the past ten years. Furthermore, a conceptual framework for how a DAO could possibly improve the situation will be designed. The findings will be presented in a research paper, video and a blockchain event that has been organized in Curaçao, in December 2018.

1.2 Boundaries

This research has a couple limitations that the reader should be aware of when reading through its contents. These are the following:

- This research will only focus only on 4 aspects of social organization (environmental, economic, political and social).
- The design for the DAO will be a theoretical design focusing only on the conceptual aspects of the DAO.
- This research will focus only on the island nation Curaçao located in the Caribbean region.
- The research has been conducted by analyzing past reports and articles only. No additional experiments will be conducted.

1.3 Reading Guide

First of all, **Section 1** is the introduction of the report. This section goes into the reason for the project and sheds light on the goals of the project and the boundaries that were considered.

Second of all **Section 2** is the theoretical background, it pertains to all of literature used in this report. The section sheds a light on the different model's organization (centralized, decentralized and decentralized autonomous). It covers the benefits and drawbacks of these methods of organization. It also looks briefly at the history of these or models of organization. Next, this section delves into the consequences of traditional centralized economic organizing by looking at shortcomings in three spheres (e.g. socio-political, socio-economic and environmental). Finally, this section also reveals information of smart contract frameworks for the development of decentralized autonomous organization.

Thirdly, **section 3** pertains to the methodology that was used for the analysis of the information gathered in the theoretical background. Furthermore, it also sheds a light on the road that was taken for the construction of the conceptual design of a decentralized autonomous organization.



Next, **section 4** delves into the results, it conveys the shortcomings that were identified and also the conceptual design that was created.

Finally, the **discussion** and **conclusion** go into the critical analysis of the findings in the results and the conclusions that were taken from the results.



2 Theoretical Background

2.1 Traditional organizing (Hierarchical & centralized decision making)

In the past centuries, the majority of human individuals or groups have been organizing to achieve goals by utilizing centralized and hierarchical organizational models (Agre, 2009). For the past centuries humans have only known two different views of organizing, the mechanistic view introduced by Frederick Taylor, Henri Fayol, Max Weber and Chester Barnard. And the human-centered view proposed by Elton Mayo, Rensis Likert, Philip Crosby, Henry Mintzberg and Peter Drucker. The mechanistic view advocates the “scientific” approach on organizing work through emphasizing economic rationality by routine, repetitive tasks, authority, lines of command, unity of control, and forms of coordination and planning. The human-centered view on the other hand advocates that employees are humans and have a crucial significance for reaching the organization's goals. Furthermore, it asserts that workers are more willing and motivated to work in a human-centered environment (Hoogervorst, Koopman, & Van Der Flier, 2005).

Even though these organizational and management views are different, when it comes to decisions making they are the same. In both views, the decision-making power within an organization lies in single or a group of individuals (board members, CEO's, Directors and shareholders) and is therefore still centralized (Hoogervorst et al., 2005). Moreover, these well-known Centralized decision-making views have been adopted by various respected global organizations such as multinational companies, militaries, communities, governments, educational institutions and many other tall hierarchical organizations in the world due to their efficiency (Blau, 1968).

Although centralized decision-making structures have been proven to be effective in making decisions. There has been literature from the past two decades which states that hierarchy compromises work performance since it reduces the motivation and innovation of working members (Bunderson, van der Vegt, Cantimur, & Rink, 2016). Furthermore, other limitations that are experienced by traditional organizations are (McDowell, Agarwal, Miller, Okamoto, & Page, 2016):

- Organizational hierarchy dictates teams
- Siloed values and culture
- Goals and projects based on hierarchy
- Information controlled by top and disseminated downward
- People rewarded for their positions instead of their work

Finally, another limitation of using a centralized decision-making model for all governance and decisions is the “nothing at stake” problem which alludes to the idea that due to the concentration of power, an individual or organization does not lose anything from bad behavior. This can open the organization up to human-engineering type attacks like discouragement, extortion, corruption or collusion (Freund & Stanko, 2018).



2.2 Decentralized organizations

While the majority of human organizations in the world are still based on traditional organizational models, there is another model that has been around since the Apache Indians (CE.1100) and these are decentralized organizations. Although there is no standardized definition for Decentralized Organizations, there have been several attempts to define Decentralized Organizations.

First, according to Brafman and Beckstrom, a decentralized organization is an organization with no clear leader, no hierarchy, and no headquarters. If and when a leader does emerge, that person has little power over others. The best that person can do to influence people is to lead by example. There are rules and norms, but these are not enforced by any one person. Rather, the power is distributed among all the people and across geographic regions (Brafman & Beckstrom, 2006).

Secondly, Buterin argues that the idea of a decentralized organization takes the same concept of an organization and decentralizes it. Instead of a hierarchical structure managed by a set of humans interacting in person and controlling property via the legal system, a decentralized organization involves a set of humans interacting with each other according to a protocol specified in code and enforced on the blockchain. A Decentralized Organization may or may not make use of the legal system for some protection of its physical property but even there, such usage is secondary (Buterin, 2014).

Thirdly, business dictionary webpage defines decentralized organizations as *“an organization wherein the decision-making authority does not sit with a central figure or group. Some decentralized organizations empower all levels within the hierarchy with decision influence.”* (Businessdictionary.com).

Finally, Vitez argues that Decentralized organizational structures often have several individuals responsible for making business decisions and running the business. Decentralized organizations rely on a team environment at different levels in the business. Individuals at each level in the business may have some autonomy to make business decisions (Vitez, 2019).

2.2.1 Reasons for decentralized organizing

The benefits of decentralized organizing are (McDowell et al., 2016):

- Teams of individuals, with individuals being part of multiple teams
- Shared values and culture
- Transparent goals and projects
- Free flow of information and feedback
- People rewarded for their skills and abilities, not positions

Other important benefits to decentralized organizing (Brafman & Beckstrom, 2006):

- when attacked, a decentralized organization tends to become even more open and decentralized.
- decentralization: is an open system that doesn't have central intelligence; the intelligence is spread.



- Can easily mutate, as soon as an outside force presents itself, the decentralized organization quickly mutates to meet the new challenge or need.
- Decentralized Organizations can grow fast.
- Participants in a Decentralized Organization/open system automatically want to contribute.
- Decentralized Organizations have a higher tolerance for innovation.

There have been recent developments aiming at strengthening decentralized organizations with the advent of the distributed systems and structures that underlie Blockchain technology. In these organizations, many aspects of a human governance are automated by code which is immutable, transparent and adds a layer of providence towards actions that are taken within an organization. This is widely known as the Decentralized Autonomous Organization (DAO).

2.3 Decentralized autonomous organizations

For many in the blockchain space, the most interesting aspect of blockchain technology is the ability to create Decentralized Autonomous Organizations (DAOs). Some define the DAOs as virtual organizations which have their rules of operations encoded in computer programs or so-called smart contracts (Chen, Stegorean, & Nistor, 2017) while others define DAOs as (Hsieh & Vergne, 2018):

“non- hierarchical organizations that perform and record routing tasks on a peer-to-peer, cryptographically secure public network, and rely on the voluntary contributions of their internal stakeholders to operate, manage, and evolve the organization through a democratic consultation process.”

However, there is a difference between these two definitions. The former definition describes an organization that operates autonomously e.g. without human involvement in governance and the decision-making process. The latter describes an organization which does involve human beings in the decision-making process, only it happens in a decentralized peer-to-peer fashion. Organizations like these are sometimes also just referred to as decentralized organizations.

2.3.1 History of Decentralized Autonomous Organizations

When one considers the definition of a DAO as explained in the paragraphs above, bitcoin falls directly under this definition. In this light, bitcoin network can be seen as a DAO, and in the traditional sense also as an organization that supports all of the four core features of an organization, namely: it is a multi-agent system (it is composed of multiple players), it has identifiable boundaries (the bitcoin ledger only stores bitcoin transactions and no other transactions), it has a purpose (the bitcoin blockchain has the purpose of maintaining a global publicly accessible ledger), and finally, the agents in the system are contributing and working towards the purpose of the network.

The minting (mining) of the first amount of bitcoin in the beginning of the year 2009 marked the existence of the bitcoin network and thus, the existence of the bitcoin DAO. Since then, many other DAOs have popped into



existence, building on the revolutionary concepts and ideas that were previously established by the bitcoin DAO. A few examples of these DAO's include Ethereum, DASH and MONERO.

DAO's can also be built as second application layers on top of existing DAOs, for example, the bitcoin blockchain. One of such developments was the Genesis DAO, which was a venture capitalist fund built as a so called "smart contract" (more on that in the following chapters) on the Ethereum blockchain. It was considered to be the first of its kind. Built by the Slock.it team, the fund, entirely autonomous and decentralized, would allow stakeholders to pitch ideas which would then be voted on by the community. The approved ideas would then be funded entirely by the DAO through the dissipation of DAO tokens. At first, it seemed the project was going to become a huge success since around 150 million dollars (12.7 million ETH at that time) was crowdfunded in the initial creation phase of the DAO, making it the largest crowdsourced project ever created. However, things turned sour when a loophole was found in the smart contract that allowed a hacker to siphon funds from the DAO. This resulted in the loss of around 3.6 million ETH which amounted to around 70 million dollars at that time. As a result of this mayhem, the Ethereum blockchain had to be forked in order to give back the stolen ETH to the investors which caused a split in the Ethereum community (Falkon, 2017).

It is important to note however, as stated before, that there is no general agreed upon definition for a DAO. In fact, many individuals named it in various ways. The different names of DAOs include: DAC (decentralized autonomous corporation), DNO (decentralized network organization), DAS (decentralized autonomous societies) and DAC (decentralized autonomous communities).

Breaking down the DAO

According to Walsch, DAO's are fundamentally egalitarian and co-operative. a DAC or DAO is an organization that unlike traditional organizations does not have its ownership represented by a board of directors. Instead, the ownership of a DAO is verified on a public distributed ledger like, for example, that of bitcoin. Also, the power to make decisions is not left in the hands of just one individual. Instead, it is distributed and spread over all of the stakeholders, those being all of the peers that form the foundation for the distributed network. He also states that even the work done, is distributed amongst peers, for example, the computer work that is required for the validation of transactions within the bitcoin network is distributed amongst all of the miners (Walsh, 2014).

Furthermore, in a DAO, there are no management layers. A lot of the management work consists of connecting supply and demand with the goal of reducing uncertainty. According to Paul Bessems, most of the actions that comprise the connection of supply and demand are divided into mechanical tasks and organizational tasks. The mechanical tasks include: planning resource distribution, control and registration subsequently the organizational tasks which are comprised of motivation and development (Bril & Bessems, 2016). He further states that:

"All of these tasks are tasks that are employed in order to reduce uncertainty when transacting with one another, where especially the mechanical tasks are for the most part automatable"



Thus, in a DAO, management is replaced by a blockchain protocol which allows stakeholders within the organization to achieve consensus and transact with one another without the need of a central third party to providing trust and thus reducing uncertainty. Participators within a DAO can all make proposals that can be either down- or upvoted by the stakeholders.

Building blocks

There are a few essential building blocks that constitute a DAO. These building blocks can be found in most if not all DAOs. Generally, they are composed of two basic elements: a **distributed cryptographically secured public ledger (blockchain)** and **smart contracts**.

Distributed cryptographically secured public ledger (blockchain)

One key component for a DAO to be able to exist is what is called a distributed ledger (i.e. Blockchain). A blockchain is the result of the addition of what are blocks of information (e.g. timestamped transactions) to an existing chain of information blocks in which each block of information references the previous block through what are so called cryptographic digests (randomized string of characters, hash) (see Figure 1).

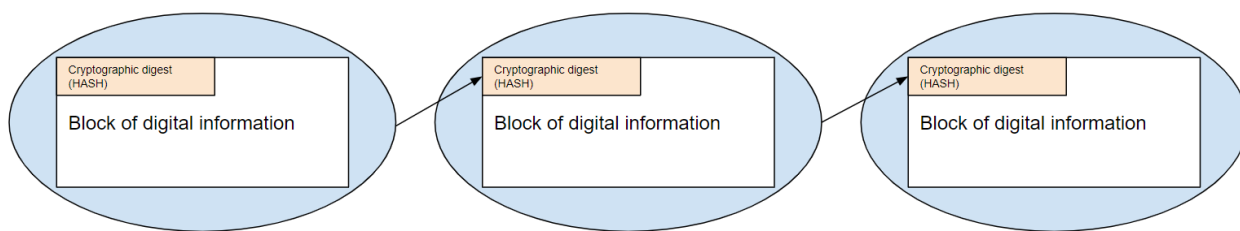


Figure 1 schematic overview of a blockchain ledger

Because of the cryptographic and distributed nature of the ledger, the ledger is deemed extremely secure and immutable by design. A block is comprised of information pertaining to the transactions that have taken place between different clients and servers on the network. Since transactions take place at different times in space it becomes a problem to register each transaction at the same time. A solution to solving this problem is what is called mining. Software installed on some servers or nodes within the network scan the network for transactions then bundle them in a block. Subsequently, the transactions are validated, and each is given a specific timestamp. After a consensus is reached by the nodes, the ledger is updated. In bitcoin blockchain for example a block is mined every 10 min. Once a block is mined, or in other words, added to the previous chain of blocks, it cannot be altered thus rendering the ledger immutable (Bril & Bessems, 2016).

There are different forms of blockchains e.g. public, private and hybrid. Public are open in such a way that anyone can connect a node to the existing network (bitcoin) and thus take part in the mining process. In private blockchains, a third party owns the network and thus has control over who gets to become a part of the network and the supply of the digital currency that powers the network (ex. ripple). In hybrid blockchains, everyone can become part of the network. However, some nodes can be programmed with the power to allow or disallow access. On another note, the security of a blockchain can only be guaranteed when 51% of the network is



controlled by all of the users. In any other scenario the probability will remain that the history of transactions can be tampered with (Ibid.).

Smart contracts

Another fundamental component of a DAO is a smart contract. A smart contract is an algorithm that takes a set of actions once a particular condition is met. Take for example the process of requiring the deeds to a house. The whole process from, searching for a particular deed to filing it (officializing) at a notary office and getting it stored in a clerk's office, can be simplified in a set of actions or steps that are taken once a certain part of the process is completed (condition is met). Alternatively, after the deed is found, file it to the notary office. Or, after the deed is signed in front of the notary and has received an official stamp from the notary, store the deed in the database at the clerk's office (Ibid.). These, set of actions can be automated and the way they are generally applied in smart contracts are with the use of what are so called "IF THIS, THEN THAT" statements which are just a set of rules that can be programmed (Ibid.).

Smart contracts allow for the reduction of costs in the preparation, execution, control and registration of transactions. This is because the execution of a smart contract is less prone to human error and manipulation. Human error and manipulation can still exist, but it then only at the hands of the programmer. Furthermore, smart contracts are also prone to attacks of hackers and errors in the code. However, once a smart contract functions properly and it is deemed secure, the chances of human error in a particular transaction process will greatly diminish. This is a function of the fact that a computer rarely makes errors in calculations, does not make mistakes and has no problem in doing large and complex calculations simultaneously. A computer has no problem remembering the conditions that lead to a sequence of actions taking place. Furthermore, it can easily keep track on whether the value of certain is met or not. Not to mention, a computer never sleeps and does not need a vacation or a pension (Ibid.).

2.4 Benefits of the DAO

Because, according to Paul Bessems, DAO's are based on technical scripts, they are in some instances (e.g. the processing transactions):

"more efficient and better able to share information and make decisions compared to organizations that are based on a social script (culture, motivation, interpretation etc.). In this sense traditional borders of a company are not heard of within a DAO. Instead only the fundamental building blocks, interconnection, instructions and conditions which are represented by a protocol"

Thus, this results in the reduction of costs related to infrastructure and organization in traditional organization. The separation of information provision between companies is complex and more difficult to subject to change than within or between DAO's. In traditional companies you have to deal more with fixed contracts, IT systems and labor agreements which are more prone to manipulation and influence compared to software code, data and data policies represented by DAO's. Also, DAO's, unlike traditional organizations, are less prone to political influence, which means that the DAO can exist unhindered provided that the integrity of the blockchain is



secured. This, again, is because of the fact that in a DAO the relationship between supplier and enquirer is seen as a connection instead of a social contract. Politicians are not able to influence software code and immutable data unless the possibility to do so is encoded within the code itself. This characteristic of the DAO allows for more security, less corruption and unhindered continuity when, for example, providing social services (e.g. social security, taxation etc.) (Ibid.).

Another benefit of DAO's is that these types of organizations are less prone to complications resulting from takeovers and mergers. What results after mergers are just a collection of new interconnections which are still driven by the same ontology. This means less costs related to information infrastructure. Next to having lower costs resulting from mergers or takeovers, DAO's are also more efficient than traditional organizations when it comes to the design, creation and adoption of new ideas because decision-making is not hindered by misplaced authority and power. All actors within a DAO can propose ideas and the whole network can vote and decide whether resources should or should not be freed in order to realize an idea. In this sense, a DAO is able to rely more on the wisdom of the crowd rather than that of a central authoritative figure (Ibid.).

Finally, working with DAO's requires the creation of new kinds of expertise that results in the creation of new working opportunities and thus the sustenance of more livelihoods (Ibid.).

2.5 The drawbacks of a DAO

One of the general ideas of DAO proponents is that the advent of technology will pave forth a new way of economic organizing in which third parties (particularly the state) will be removed all together. This removal of third parties will allow for freedom and liberty of trade between networks, creating new forms of markets where, as cited by Garrod (Garrod, 2016):

“one where users are in control, not governments or big companies”

There are some however, whom seem to disagree with this idea and instead argue that this idea is not necessarily true and that the creation of decentralized autonomous societies (DAS) - basically societies that are governed by autonomous institutions with no human centralized control and power - could actually lead to more consolidation of power and control. The argumentation further continues that (Ibid.):

“While there are many grand claims, there seems to be little understanding of the wider social context in which DAS are embedded.”

As argued by J.Z. Garrod, in his essay: “The real world of decentralized autonomous societies (DAS)”. The discussion of DAS disregards many important socio-economic factors that underlie the validity of societies, mainly: class and the role of the state, the concentration of capital and automation (Ibid.).

2.5.1 Class and role of the state

One of the arguments that Garrod makes is that proponents of DAOs or DAS, who go around advocating the idea that removal of the centralized third parties (e.g. the state) will bring more liberty and freedom actually, lack a



basic understanding of the history of how those institutions came to exist in the first place. In his view, DAS proponents, in their different discussions, base their view on the commercialization model, which premises that capitalism is the inevitable result of human nature. He goes on further stating (Ibid.):

“However, this line of thinking ignores the actual history of the transition toward a capitalist society, also called "primitive-accumulation". A time period of time in which emerging capitalists used the power of the state to transform feudal rationality and introduce a capitalist one. This same line of thinking was also, in more recent times, utilized by neoliberal thinkers to justify privatization, open markets and deregulation. For example, neo-liberal thinker Milton Friedman premised that the problem of social organization can only be resolved through competitive capitalism. However, this narrative does not consider the coercion involved in the separation of capital from labor or the possible mitigation of it by the regulatory and welfare state.”

Garrod further argues that in entertaining this economic worldview of DAS proponents which is based on the premise that crony capitalism is consequential to the existence of the government and the state (so-called centralized third parties). The actual role of securing conditions for capital to exist and the historical defense of wealth is ignored. He goes on to propound that by ignoring this role the future of a stateless capitalist society can be imagined. However, in actuality this future would be, as Garrod writes (Ibid.):

“characterized by extreme inequality, poverty, and private authority resting on the ownership of capital: a form of “distributed capitalism” as Kostakis and Giotitsas (2014) put it, in which peer-to-peer infrastructures are utilized to accumulate and secure capital”

Regarding class relations, Garrod argues that the scenario could be more oppressive than ever in a stateless capitalist society governed by DAOs. Not everyone will be able to code the DAOs, only an emerging technocrat class. Also, according to Garrod, there is little discussion about how material property (e.g. land) relates to the “absolute sovereignty over resources” inherent to DAO’s as proclaimed by DAO proponents. He argues that in his scenario of DAS, already existing third parties (e.g. states and other private authorities) will be used to secure material property and with regard to digital property which compose of various financial forms including debt the “techno-leviathan” (DAS) will be used to protect those. What this could look like is the use of software that are used to enforce, control or restrict the access to physical and digital property. Said access would be regulated by the technocratic class which have the power to create these new forms of institutions, generally taking away what used to first be done through law and regulations enforced by people. And replacing it with algorithms that mathematically enforce these laws (Ibid.).



2.5.2 Concentration and centralization of capital

Another potential problem of DAS mentioned by Garrod is the problem of centralization of capital. Garrod argues that if the adoption of DAS is followed by guided by an economic trend of capital accumulation through the removal global governmental trade barriers in the name economic growth (Ibid.):

“There will be little prospects for protecting any sort of commons”

that capital tends to concentrate and centralize. In the context of capitalist competition, this will result in the emergence of increasingly high levels privatized capital. This idea is explained by looking at the way capital has become concentrated and centralized over time, as Garrod notes (Ibid.):

“Through the early period of capitalist development, through the rise of imperialism and monopoly capital in the late nineteenth century, capital has become increasingly concentrated and centralized, existing today in the form of the transnational corporation (TNC)”

In recent times, the growth, concentration and consolidation of capital has materialized itself in the form of non-nation state forms that are there to protect the rights of capital over the rights of jurisdictions through the creation of many different trade agreements (CAFTA, NAFTA, WTO, TPP etc.). These agreements have the function of freeing up capital from the limitations that laws and regulations from different governments. In other words, they allow TNC's to sue governments if they negatively impact their accumulation of capital. Thus, as Garrod notes, it is not just about whether or not blockchain technology works. It is about the way the technology is applied. Chances always remain that, as Garrod writes (Ibid.):

“property relations become increasingly denationalized in such a way as to allow the private authority of capital to reign across the globe”

2.5.3 Automation

Another issue that Garrod attempts to address in his critique DAOs is the issue of automation. Garrod notes that amongst the proponents of DAO's there exist this belief that DAS will usher a new post-capitalist era where means of production are fully automated and where, as he cites, the pillars of societies are represented by “a collaborative common in which everything is open source”. However, as Garrod further argues, given the current capitalist trend, the process of adopting blockchain technology might not lead to ideal manifestations of societies as proposed by blockchain proponents (Ibid.).

Garrods explains that, the logic behind the views of DAS proponents is based on the idea that with an increase in automation and thus the increase in productivity, the marginal cost (cost of producing additional units) will decrease to zero. And since profit is made at the point of marginal cost, this means that at some point this could lead to a situation where products and services will be made freely available and shared in collaborative networks just like is already happening in several media industries (Ibid.).

He further continues that the problem with this idea is that the form of profit making has transformed in contemporary capitalism. Nowadays corporations secure profit by ownership, a form of rent, realized through (Ibid.):



“valueless knowledge commodities, which are “privatized ideas, commodified knowledge, know-how, information, and instructions” that only employ labor for first-time production, and not continued re-production.”

When considering this issue, the idea that zero-marginal cost will exist because of automation becomes highly questionable. Moreover, if the monopolies of large transnational corporations remain intact. As Garrod further cites (Ibid.):

“the main impetus behind the ramped-up industrialization [...] lies not so much with the consumer, but the potential benefits to multinational industrialists that will be its earliest adopters”

He also notes that it is common for DAS proponents to take the view of Keynes, in which he argued that automation would free us and thus allow us devote time into non-economic purposes (reference). However, the reality shows us, as Garrod writes, that since the time of Keynes (Ibid.):

“people work more than ever; automation has not lessened our working day, but rather, increased it. And furthermore, the increasing use of this technology in the production and service sectors means greater unemployment for the majority of the people on earth “

On a final note, Garrod argues that even with all of the potential of blockchain technology, it is not yet clear (or discussed thoroughly) how the technology will help free the subordinate classes. What seems to be the more likely outcome is that the technology will chain said classes even tighter to the grip of capital subjecting them to new form of rentier activity (Ibid.).

2.6 Smart contract frameworks for DAO's

It is becoming increasingly easier for individuals to build DAO's. This is because there is a lot of development going on with regard to the creation of DAO's e.g. the development of what are so called frameworks for building DAO's. Generally, these frameworks consist of prewritten chunks of code (e.g. smart contracts) that have a particular function within a DAO. These frameworks can also be compared to the multitude of libraries and frameworks utilized by developers for purposes of developing different products and services (e.g. apps, websites, software) on and off the web.

With this new approach of using fundamentally decentralized programmable protocols with the ability to design and create efficient, decentralized, transparent and frictionless human organizations, it now becomes possible to code any type of organization with any form of governance that is desired. The number of smart contract frameworks are increasing by the day, to mention a few: Colony, Aragon Network and Daostack's Arc.

2.6.1 Colony

Colony is a new kind of internet-native digital company or organization where rules are enforced by software instead of hierarchy. Colony stands for reducing the organizational middle management, administration and authority so fewer people can get more done. Furthermore, Colony provides the opportunity to create modular and customizable digital organizations. These can vary from benevolent dictatorship to democratic cooperative.



Everyone can create an organization on Colony. However, like many other projects important to understand what type of organizational technology Colony fundamentally relies on (Colony, 2019; Rea, Fischer, & Rose du, 2019).

Infrastructure underlying Colony

- **Ethereum** - Ethereum is a global, open-source platform for decentralized applications: public blockchain infrastructure for software developers to write code that can control digital value, runs exactly as programmed, and is accessible anywhere in the world.
- **IPFS** - IPFS is a distributed system for storing and accessing files, websites, applications, and data. Being distributed makes it hard for websites to go offline, hard for authorities to censor content, and means no central entity has to provide storage.
- **OrbitDB** - Orbit-db is a serverless, distributed, peer-to-peer database. Orbit-db uses IPFS as its data storage and IPFS Pubsub to automatically sync databases with peers. It's an eventually consistent database that uses CRDTs for conflict-free database merges making Orbit-db an excellent choice for offline-first applications.

The Colony Network

Represents a suite of smart contracts that run on the Ethereum blockchain. It provides a general-purpose framework for the essential functions' organizations require, such as ownership, structure, authority, and financial management.

The Colony Network has various capabilities and therefore use-case agnostic, enabling organizations to customize their Colony with smart contracts modules providing the rules their organizations require. These capabilities are (Colony, 2019; Rea et al., 2019):

- **Payments** - From salaries to gigs, and bounties to tips, payments can be denominated in Ether, DAI, or any ERC20 compatible token.
- **Domains** - An organization can structure themselves into teams, circles, projects, or whatever is appropriate.
- **Budgeting** - An organization's Domain can allocate & manage funds via a given set of rules.
- **Reputation** - This will reflect members' work, skills and achievements of organizations built on Colony. A built-in social crediting system for organization members.
- **Permissions** - The distribution of decision-making power will be possible and can be assigned to people, bots, or smart contracts.
- **Disputes** - Raise objections, resolve disagreements and incentivize a colony (organization) to self-regulate.
- **Contract execution** - A colony may execute external smart contract calls, enabling plug and play governance for Ethereum protocols.



Metacolony

As the Colony Network can be viewed as a self-sustaining public utility: “a digital commons available to all, in which permission isn't needed and censorship not possible.”

In its infancy the infrastructure requires upkeep (ongoing management and development). This is to support the growth of a thriving Colony DAO ecosystem. For this reason, a Metacolony has been created where the focus is on important work. As stated on the colony.io website “every colony needs the Metacolony”. This can be seen as the main colony DAO that feeds the Colony ecosystem. In the Metacolony everyone can contribute and will be launched under the aegis of Colony Foundation Ltd., which will gradually cede control to the Metacolony’s members.

Here the fees that are required to executed transactions on the Colony platform will go to the Metacolony to sustain ongoing development, security and Colony’s ecosystem incentive mechanism: [BudgetBox](#) (Colony, 2019).

Some detailed information regarding Metacolony’s financial sustainability:

The Colony Network levies a 2.77% fee on all Payments from each Colony (DAO). Fees paid in ETH (Ethereum token symbol) or [DAI](#) (MakerDAO stable coin token symbol) go to the Metacolony. Fees paid in other ERC20 tokens go to auctions. There, token buyers can purchase ERC20 tokens (tokens created on the Ethereum blockchain) using Colony's Native Token, CLNY. The CLNY is then burned (Colony, 2019).

The roles of CLNY token are the following:

- Governance
- Reputation mining
- Revenue sharing
- Network fee auctions

Structure within a Colony (organization)

Within Colonies effective division of labor is, one of the most important functions of the Colony protocol. Work is divided into discrete ‘tasks’, which can then be organized into ‘domains’.

The smallest structural unit in a colony is the ‘task’. This represents a unit of work requiring no further subdivision or delegation. A task has three roles associated with it:

- A manager – someone responsible for defining and coordinating the delivery of the task.
- A worker – someone responsible for executing the task
- An evaluator – someone responsible for assessing whether the work has been completed satisfactorily.

These roles are assigned to addresses. While it is anticipated that roles within tasks will likely be assigned to individuals, there is nothing to prevent these addresses being contracts under the control of multiple people.



A domain is like a folder in a shared filesystem (see Figure 2), except instead of containing files and folders, it can contain sub-domains and tasks. This simple modularity enables great flexibility as to how an organization may be structured. An example is shown in the following figure:

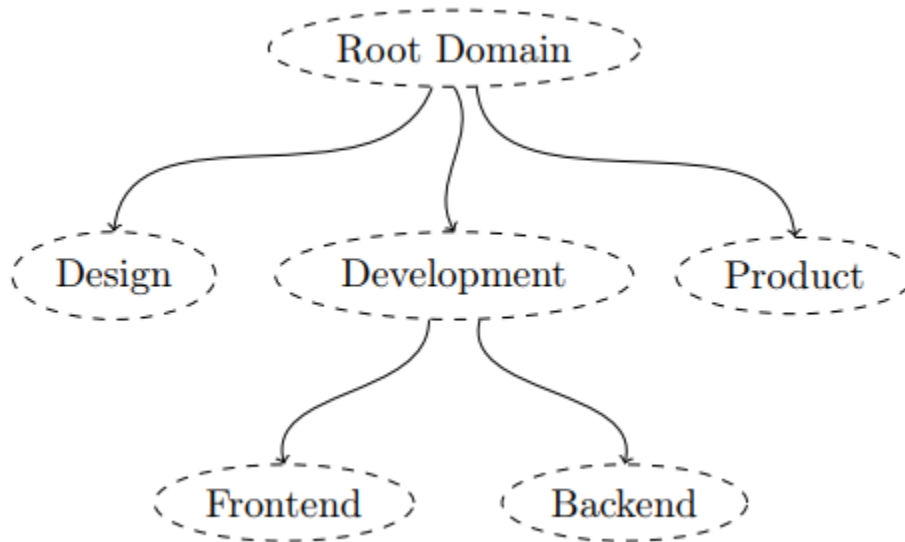


Figure 2 A Domain structure within colony (Rea et al., 2019)

This compartmentalization of activity provides a tangible benefit to the colony by making reputation contextual. When objections are raised, they are raised at a specific level in the colony's domain hierarchy. This means that people with relevant contextual knowledge can be targeted for their opinion, and that when a dispute occurs, the whole colony is not required to vote in the dispute. Rather, only members with the relevant experience can provide their opinion. However, it is ultimately up to individual colonies to decide how they wish to use domains (Rea et al., 2019).

Lastly, in a recent (27/10/2019) Discord conversation with Pat from *the MetaColony* he mentions more about the differences between Colony and other DAO frameworks: "Colony, unlike some of the other DAO frameworks, is primarily focused on getting day to day work done. It provides the framework for individuals to coordinate around tasks and tools to govern themselves when things go wrong. Other DAO frameworks seem to focus on voting-based actions. While can be effective at coordinating big decisions, they become bureaucratic for day to day work. Colony wants to stay out of the way until needed."

2.6.2 DAOstack's Arc

DAOstack's Arc enables organizations with dozens, hundreds and even thousands of members to make collaborative decisions on for example, how to spend funds in alignment with the organization's goals and values. DAOstack's Arc is designed from the ground up to be very modular and pluggable and can therefore be used for an array of organizational structures & rules. The core infrastructure services DAOstack's generic platform Arc provides (DAOstack, 2018):



- **Controller** - The controller is not owned by anyone and gets commands from the schemes alone, which operate its functions. Through these functions, it sends commands to the “actors” contracts, as well as upgrading functions of the governance system and the technical architecture itself.
- **Agent inputs** - The only interface for agents to interact with the agency is via its schemes. Each scheme comes with a specific “knob” which can be operated by external agents (Blockchain addresses) through transactions that call and operate these functions.
- **Actors:**
 - **Token printer** - The token issues and allocates the native tokens of the agency to agents. It is owned by and receives commands from the controller alone.
 - **Reputation printer** - Assigns the agency reputation to agents. It is owned by and gets commands from the controller alone.
 - **Avatar** - the face of the agency (and its address), which acts externally and is capable of doing - through the help of particular schemes - anything that can be done on the blockchain. For example, it can participate as an agent (as a blockchain address) in other agencies (DAO’s) - e.g. submit and vote on proposals - on behalf of its agency. It can also be the identity to which reputation in other agencies is assigned. In short, the Avatar has only a few functions, and they can only be involved by a Controller.
 - **Funds (Wallet)** - It holds external funds owned by the agency and distributes them to agents. It is owned by and gets commands from the controller alone.
- **Universal elements:**
 - **Schemes** - simple pieces of logic that comprise the different actions, that can be executed by the controller. One example is an ICO scheme whereby an agent who sends Ether to an agency and receives tokens of the organization in return.
 - **Constraints** - constraints or global constraints are forced when considering modular logic as one generally wants to prevent the modules from breaking certain overarching rules. An example is a cap on an organization’s total possible reputation, or a maximum burn rate of an organization’s funds.
- **Upgrade:**
 - **Protocol upgrade** - A controller can when receiving commands from the subscribed schemes - register new schemes and constraints or unregister old ones. Since the governance protocol of an agency is fully specified via its schemes, every agency can easily upgrade and modify its governance system (if this ability is rooted in its existing protocol).
 - **Technical upgrade** - Special function of the controller that transfers its ownership over all of the actor contracts to a new address. Whatever would be this new address, it will have a complete control over the assets (transferable and non-transferable, such as reputation in other agencies) of the agency. This can be an upgrade to an improved architecture of Arc, or a whole new architecture, if desired by the agency.

2.6.3 Aragon

Aragon is a tool for creating and governing organizations such as clubs, companies, gaming guilds, cooperatives, nonprofits, open source projects, and any other type of organization you can imagine. Aragon currently supports



creating organizations on the main Ethereum network (“mainnet”) and the Rinkeby test network (“testnet”). The two main underlying components of Aragon DAOs are the AragonOS framework and the Aragon client.

AragonOS

AragonOS is the smart contract framework of Aragon for building decentralized organizations, decentralized applications, and protocols. The OS in AragonOS stands for *Operating System*, the same way a computer operating system manages how applications access the underlying resources of the computer—the hardware AragonOS does this for decentralized organizations or protocols. Its architecture is based on the idea of a decentralized organization or protocol being the aggregate of multiple components (called applications) connected by a pillar, named the Kernel, which are all governed by a special *Access Control List* (ACL) application that controls how these applications and other entities can interact with each other.

AragonOS provides the following functionalities:

- **Upgradability:** Smart contracts can be upgraded to a newer version. Example: fixing a bug or adding a feature.
- **Permission control:** By using the `auth()` and `authP()` modifiers you can protect functionality such that they're only accessible by other apps or entities if given permission. This completely abstracts the authentication logic from an app, allowing you to focus on your app's business logic. Example: protecting a vault so only the organization's Voting app can initiate an action to transfer funds.
- **Forwarders:** aragonOS apps can communicate with each other by sending their intent to perform an action to other apps. Example: withdrawing funds from a vault only on the passing of a vote and the expiring of a time-lock.

All the above makes it very simple for aragonOS apps to incorporate governance. On top of AragonOS there is the Aragon client which mainly exists out of the front end (app UI & wrapper UI)(Aragon, 2018b, 2019b).

Aragon client

The upper component of Aragon DAOs is the Aragon client which is the UI framework which is built to be Aragon native. This means that anybody in the world can easily take components out of this and make applications on the front end that look very similar to the decentralized applications Aragon currently deploys (Aragon, 2019c, 2019d).

Most organizations that use Aragon to build their DAOs will be created through an Aragon Template. Currently at the time of writing with Aragon 0.8, the client has the following templates:

- **Company:** A “Company” organization that uses transferable tokens to represent stake in the company. Decisions are made using token-weighted voting, where one token equals one vote.
- **Company-Board:** The Company with Board template is intended to be used as the basis for a token holder governed organization that has deferred some of its decision-making power to an elected committee of “board” members. The “share” tokens are freely trade-able and can be added to



exchanges. The "board" is represented by non-transferable tokens that are limited to one token per address.

- **Membership:** A "Membership" organization that uses non-transferable tokens to represent stake in the company. Decisions are made using token-weighted voting, where one token equals one vote. Token holders in a Membership organization can only hold one token at a time, making the voting style "one-member-one-vote".
- **Reputation:** A "Reputation" organization that uses non-transferable tokens to represent reputation. Decisions are made using reputation-weighted voting, where one reputation token equals one vote.

Aragon organizations could extend their functionality using Aragon apps. When a DAO is created using the Aragon client's above mentioned templates, the organization will include four pre-installed apps (Aragon, 2019a).

- **Token Manager app:** allows to mint/burn an organization's token and to assign/remove them to addresses.
- **Voting:** allows members in an organization to vote on a question-which may then lead to the execution of an action based on the outcome.
- **Finance:** allows members to manage the organization's financial capital.
- **Agent (optional):** The Agent app enables Aragon organizations to interact directly with any other smart contract on Ethereum.

A simplified visualization of how the Aragon framework layers look is presented below (see Figure 3):

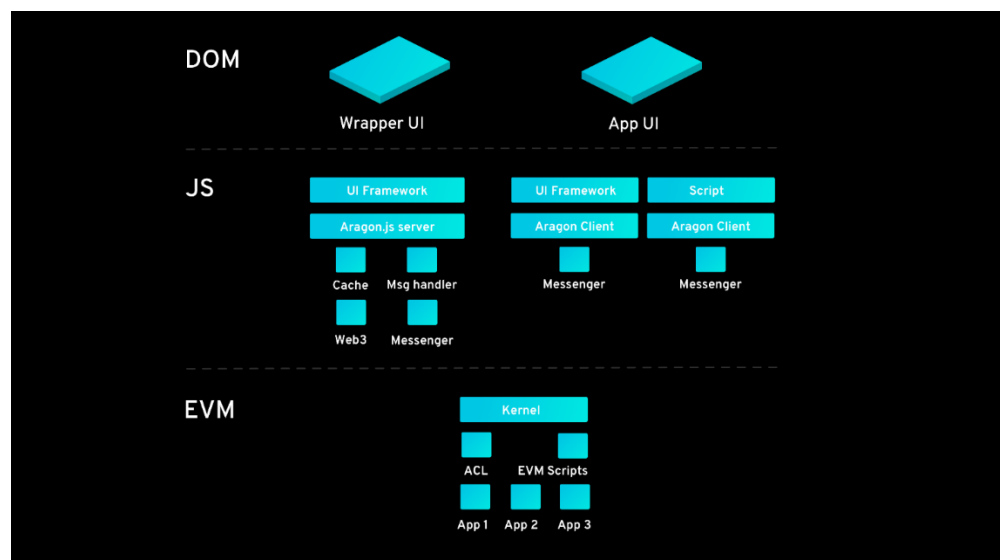


Figure 3 Conceptual diagrams about the architecture of the Aragon client and AragonOS (Aragon, 2018a)

In this figure the EVM (Ethereum Virtual Machine) & JS (JavaScript) layers are what the AragonOS exists out of and the DOM layer is what is seen as the Aragon client. Anyone that wants to build on or with Aragon software would need to understand these different components.



Important Aragon organizations that also will or are making use of this framework are the Aragon Network & the Aragon Court. In a recent (14/10/2019) conversation with John Light (Communications lead of Aragon One) from Aragon One he mentions more about the Aragon Court & Network:

"The Aragon Court is a smart contract system that Aragon organizations will be able to use to resolve disputes. The interface organizations will most likely use to interact with the court will be a "Proposal Agreements" app that organizations can install and access using the Aragon client.

The Aragon Network is the DAO that oversees the Aragon project. This DAO is governed by ANT holders and is responsible for managing the assets of the project (such as our treasury) and official policies of the project (such as the community code of conduct, GitHub repo management, etc.). When the Aragon Court goes live the Aragon Network will also govern the smart contract parameters of the court.

The Aragon Network does already use the Aragon client for its own governance such as the [Aragon project](#) as well as several "sub-DAOs" used for different purposes for example, [AGP 10 DAO](#) & [Aragon events DAO](#)"

2.7 Consequences of economic organizing on Curaçao

Similarly, to other economies in the world, economic organizing in Curaçao is realized through centralized decisions making based on hierarchy. There are three sections (spheres) in which economic organizing is realized, mainly: the **socio-political sphere**, the **socio-economic sphere** and the **socio-environmental**. The socio-political sphere refers to all of the institutions that coordinate themselves to achieve economic organizing in the political spectrum and their influence on society. Next, the socio-economic sphere refers to all of the institutions that are involved in the economic organizing within the economic spectrum (primarily business) and their influence on society. Finally, the environmental sphere involves all of the institutions within the environmental sector involved in economic organizing and their effect on the society.

2.7.1 Socio Political sphere

Governance in Curaçao

As an island nation with a long-recorded history with the European country the Netherlands, Curaçao's governance framework is for the most part derived from the Dutch system. The governance structure of Curaçao, a "constituent country" (Dutch: land) of the Kingdom of the Netherlands, takes place in a framework of a parliamentary representative democratic country, whereby the prime minister is the head of government, and of a multi-party system. Executive power is exercised by the government. Legislative power is vested in both the government and parliament. The judiciary is independent of the executive and the legislature.

Curaçao's political system and its characteristics can be identified as a democratic system where representatives are elected through elections every four years and enter the parliament. Politicians can make themselves eligible by postulating themselves through affiliation with a political party. When a cabinet is formed based on the majority support of the elected parliament (coalition), these political parties respectively choose who, amongst them or their peers, are to be appointed as ministers.



Political drawbacks

Humans generally do not trust each other with the exception of people within their personal exchange network (Chakraborty, November 26). As a result, humans have developed institutions that helped remediate this lack of trust. Curaçao, as many other countries has also developed organizations that facilitate trust within the financial, governmental, transport and other service sectors. Nonetheless, the national integrity of these institutions was brought into question in the National Integrity System (NIS) assessment on Curaçao conducted between 2012 and 2013, by Transparency International (TI).

The NIS of Curaçao is comprised of 14 pillars which represent all of the key public and private institutions that represent the governance system of Curaçao (see Table 1). In their report on the NIS of Curaçao, Transparency International assessed these pillars along the dimensions of:

- **Capacity** - this pertains overall capacity of the institution with regard to resource availability and independence
- **Internal Governance** - this pertains to the internal governance of the institutions, whether they are transparent, accountable and act with integrity
- **Role within Governance system** - this pertain to the role of the institutions within the overall integrity system. The focus is on the degree to which institutions in this pillar fulfill their roles with regards to preventing and fighting corruption.

Table 1 List of institutions in the National Integrity System in Curaçao (Transparency International, 2011)

Core Governance Institutions	Public Sector Agencies	Non-Governmental Actors
Legislature	Public Sector	Anti-corruption agency
Executive	Law enforcement agencies	Media
Judiciary	Electoral management body	Civil society
	Ombudsman	Political parties
	Supreme audit and supervisory institutions (public sector)	Business
	Supervisory institutions private sector	

Each dimension was measured by a set of indicators with the goal of examining both the legal framework and the actual institutional practice. The table below gives an idea of the matrix that was used in the assessment by NIS (see Table 2). Each indicator was given a score out of maximum 100 points. With 81-100 meaning very strong, 61-80 meaning strong, moderate was 41-60, weak was 21-40 and very weak was 0-20 (International, 2011).



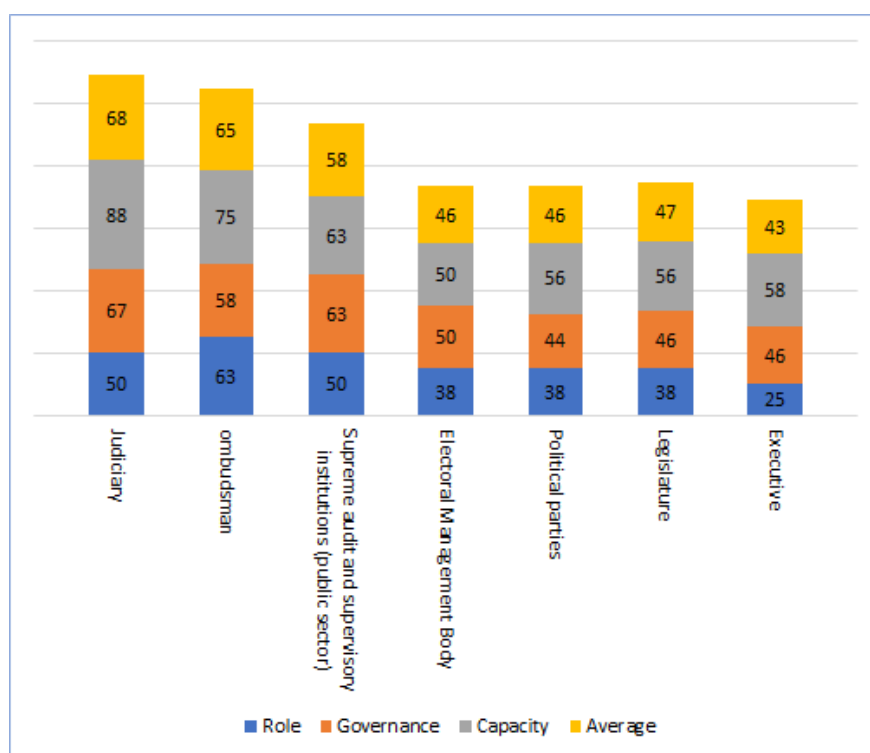
Table 2 NIS indicators (Transparency International, 2011)

Dimension	Indicators (law and practice)
Capacity	Resources
	Independence
Governance	Transparency
	Accountability
	Integrity
Role within Governance system	Pillar specific indicator

The NIS assessment estimated that the strongest (scored the highest) institutions within the NIS of Curaçao are the judiciary, the Ombudsman and the supreme audit and supervisory institutions (see Table 3). However, their strength was found to be undermined by the weaknesses in other pillars of the NIS namely: the electoral management body, the political parties, the legislature and the executive. Also, these pillars are not weak on their own accord, instead it is the weakness of one institution that undermines the other and reduces its strength, kind of like a chain reaction (Ibid.).



Table 3 NIS scores for each NIS institution in Curaçao (Transparency International, 2011)



Judiciary

Firstly, the assessment revealed that judiciary had an overall score of around 68 (strong). Where the judiciary scored the lowest was for the indicators related to **accountability (law and practice)** and for the **role specific (executive oversight, corruption prosecution)** (Transparency International, 2011).

With regards to accountability (law), it seemed, according to the NIS assessment, that even though judges in Curaçao are required by law to give explanations (reasons) for their decisions and that in spite of the fact that the legal framework of Curaçao contained comprehensive provisions with regard to complaints and disciplinary sanctions, there is no independent body available to investigate and also there is no protection for complaints. Also, even though members of the judiciary are required by law to report about and be answerable for their actions, the provisions are not always applied. Furthermore, while the judiciary has sufficient power to oversee the actions of the government, there seems to be a lack of enforcement of these legal powers (Ibid.).

When it came to the assessment of corruption prosecution, it was found that due to the mission of the judiciary (which is to remain an independent and impartial entity), it could only be reactive instead of proactive, with regards to the fight against corruption. Thus, it seemed that the judiciary is not always successful in penalizing corruption related offenders (Ibid.).

Ombudsman

Secondly, the assessment of the **ombudsman** was concluded with an overall score of 65 (which is considered strong). When it comes to the scores of individual indicators, it was found that the ombudsman scored



moderately for: **resources (practice)**, **transparency (practice)**, **accountability (law and practice)**, **Integrity (practice)** and **role specific indicator** (Ibid.).

From the assessment, it seems that even though generally the ombudsman receives adequate resources to maintain its functioning, recent budget cuts and procedural drawbacks make it difficult for it to complete its tasks. Furthermore, they also have difficulty with attracting and keeping adequately equipped personnel (Ibid.).

When it comes to transparency, the assessment revealed that while this institution was proactive in making information available on the organization and the way it functions, there is still more that could be done on transparency with regards to individual investigations carried out by them (Ibid.).

Regarding the assessment of accountability, it was concluded that while there are some provisions in place that require to the ombudsman to be accountable for its actions, what is still misses is more specificity regarding those provisions. Furthermore, as per the provisions, it is not mandatory for the ombudsman to have a complaints procedure. However, the assessment revealed that they do have an internal complaints procedure. Also, the contents and findings of the annual reports of the ombudsman are not discussed in parliament. Next to that, the public makes very little use of the power they have to ask the ombudsman for the information (ex. annual reports) that they are required to give according to the provisions (Ibid.).

Next, it was found in the assessment that the ombudsman does undertake some actions in order to ensure their integrity. However, these actions have only been adopted recently and thus, there was not sufficient information available to assess whether those mechanisms are actually effective or not (Ibid.).

Finally, when it comes to the effectiveness of the investigative actions of the ombudsman, it was revealed that, on a case by case basis, the ombudsman appeared to be effective. However, it was found that there is lack of cooperation between the administrative bodies (government) and the ombudsman. Further complicating the situation is the fact that there are no provisions in place that sanction (punish) non-compliance (Ibid.).

Supreme audit and supervisory sector

The assessment of the supreme audit and supervisory institutions was concluded with an overall score of 58 (strong). The lowest scoring indicators were **resources (law and practice)**, **transparency (law and practice)** and **role specific indicators (overall score of 50)** (Ibid.).

First of all, it was found that these institutions had the capacity to be responsible for their own budget and management. However, even though these institutions can acquire a budget that is sufficient enough for them to maintain their practices, the budgeting proposals were not always approved (Ibid.).

Secondly, when it comes to transparency, the assessment showed that none out of the three institutions that were studied were required by law to make their assessments and recommendations accessible to the public. However, some provisions do exist that allow the public to obtain information from the auditory institutions. Furthermore, one of the institutions (CFT) was required by law to inform the government of Curaçao (parliament) about its activities. Also, some of that information was required to be made public in the budgetary



process. In terms of practice, the assessment revealed that only one out of the three institutions published information of its activities regularly. One of these institutions does publish its reports; however, this is done in an untimely manner. Furthermore, one of the auditory institutions does not publish any of its findings at all, it does however publish annual reports (Ibid.).

Thirdly, while there exist provisions for the insurance of accountability with regard to the auditory institutions, these provisions are not always completely applied in practice, thus rendering those provisions ineffective for one of the institutions (e.g. ARC). On the other hand, when it comes to the other two institutions, the provisions are implemented effectively (Ibid.).

Finally, with regard to the score of the indicators pertaining to the role dimension, the assessment showed that when it comes to the law and practice of effective financial audits, the effectiveness of audits on public expenditure remains limited as a consequence of delayed or missing information from the public companies and public foundations in question. This is largely the result of there not being any legal requirements (or provisions) for the public sector (e.g. public companies and foundations) to make the information necessary for the audits available. Furthermore, the assessment showed that while the auditory institutions do have legal powers to detect and investigate misconduct and misbehaviors of (elected) officials and public officeholders, they do not conduct these investigations proactively or regularly. Finally, the assessment also showed that when it comes to the improvement of financial management. The auditory do have the power to note or create improvements. However, the follow up on the implementation of these improvements are lacking and thus render these improvements ineffective (Ibid.).

Electoral Management Body

Another institution that was assessed by TI was the electoral management body. The assessment concluded the integrity of this institution with an overall score of 46 (weak). With the institution scoring the lowest for the indicators relating to **resources (law)**, **independence (law)**, **transparency (law and practice)**, **accountability (law and practice)**, **integrity (law and practice)** and one of **role specific indicators** e.g. **campaign regulation (law and practice)** (Ibid.).

First of all, regarding the assessment on the resources indicator (law), the NIS assessment showed that there are no regulations or ordinances that allow the electoral management body to have sufficient resources for it to function. Instead, what is in place is a memorandum that points out that the minister is responsible for adequate funding of this institution. Despite of this, there are no concrete legal provisions that allow for the enforcement of this task. Furthermore, the assessment also pointed out that the electoral management body (e.g. electoral council) was recently tasked with acting as the supervisor for political party financing. However, the ordinance did not provide any concrete legal provisions for ensuring the covering of costs related to this task (Ibid.).

Secondly, NIS assessment revealed that there are no regulations or laws in place to assure the independence of the electoral council. Instead of being anchored in the constitution, the position of the electoral council is recorded in the electoral code. While the division of responsibilities of the electoral council is clearly elaborated in the electoral code, the code does not provide any safeguards that allow for the council's independence. For example, the appointment, suspension or dismissal of council members occurs only at the discretion of the



minister. Furthermore, the minister can appoint members of the electoral council at least 30 days before nomination day. What this means is that the legal framework allows the outgoing cabinet to exert influence over the electoral management body. Also, the legal framework does not provide clear requirements for the recruitment of professionally adequate personnel. In the same way, no specific reasons for suspensions or dismissals are stipulated clearly in the electoral code. This allows for the recruitment of inadequate personnel and thus increases the probability of favoritism within the standing cabinet. Similarly, council members can be unjustifiably suspended or dismissed by the cabinet (Ibid.).

Thirdly, the NIS assessment also noted that when it comes to transparency, while the electoral code does provide many legal provisions that ensure the transparency of the electoral council, further improvements remain possible. Specifically, introducing a provision that requires the electoral council to make all of its formal reports public. On the other hand, it was revealed that there are no legal provisions in place that assure the transparency of the council's work related to political party financing. Additionally, with regard to the actual practice of transparency by the electoral council, it was revealed that while the council was transparent about all of its activities related to electoral process. When it came to its activities related to political party financing, the electoral council did not make any of that kind of information available to the public (Ibid.).

Next, the NIS assessment revealed that regarding accountability, while there are several provisions in place which hold the electoral council accountable for its actions related to the political process. The body is not required to present comprehensive report once the elections are over. Also, there are no provisions available to hold the electoral council accountable for executions of the its tasks related to political party financing. In practice, it did seem to be the case that the electoral council was actively providing formal reports to the Ministry of Administration, Planning and Service. Furthermore, there were explanations (redress) for any irregularities during the electoral process. However, it was found that with regard to its activities on political party financing, the body did not report any of them (Ibid.).

Also, the NIS assessment of Curaçao also assessed the integrity of the electoral management body. It found that while the electoral code does include integrity rules that have to be applied by civil servants in the electoral process, there are no rules to ensure the integrity of appointed council members or those appointed to the polling stations. Secondly, with regard to practice, the NIS assessment showed that as a result of the lack of a comprehensive approach to ensure the integrity of the electoral council, and that of the supporting polling stations, it was difficult to assess whether the integrity of this body was ensured in practice. Moreover, the NIS assessment also mentioned that the integrity of the body has never before been questioned (Ibid.).

Finally, with regard to the assessment of the role specific indicators, the NIS assessment showed that there was no effective regulation of campaign finances of political groups in practice. This is the result of the new ordinance with regard to the auditing of campaign finances of political parties having not yet been enforced by the body (Ibid.).

Political parties

The assessment of the NIS of Curaçao also looked at the integrity of political parties. In their assessment, the TI gave the political parties an overall score of 46 (weak). With the indicators relating to **resources (practice)**,



independence (law and practice), transparency (law and practice), Accountability (law and practice), Integrity (practice) and all of the role specific indicators (Ibid.).

Notably, when regarding the resources indicators the assessment presented that while there are legal provisions in place which allow for the creation of an environment that is “conducive to the formation and operation of political parties”, in reality it is not known what kind of financial resources are available. Thus, TI (Transparency International) could not determine to what extent the available resources allow for effective political competition. Furthermore, the situation becomes even more difficult to assess since there are no legal provisions that obligate political parties to make their finances public. In the assessment, TI did suggest however, from the observations they had made, that there was: “an increasing importance of and dependence on private financial donors”. Also, the TI showed that ruling political parties have an unfair advantage over upcoming parties since they are allowed to use state resources for political purposes (Ibid.).

Secondly, regarding the assessment of independence, it was found that, while there exist some legal provisions that allow for the functioning of parties without outside interference, there are some important loopholes with regard to regulation of party finances that need to be addressed. The possibility still exists for parties to receive funds through donations from the private sector and public foundations. In practice, no indication was found of state attempts at dissolution or prohibition of political parties. It could not be assessed what the influence of party financiers were on political parties, since information on party financing was not publicly available (Ibid.).

Thirdly, the assessment showed that with regard to transparency, the law did not require the political parties to be transparent about their finances. It was also revealed that in practice, political parties were not transparent about their financial practices nor did they reveal the sources of their finances to the public (Ibid.).

Regarding accountability, the assessment revealed that while new provisions have been introduced, loopholes exist in the current legal framework that hinder effective oversight of political parties by the electoral management body. The new provisions obligate political parties to send information on their finances to the appointed third party (i.e. the electoral council). However, some finances (e.g. money received through fundraising activities) are not required to be registered in the same way as items such as gifts. Furthermore, only gifts above a certain threshold are required to be registered (e.g. \$2800). Therefore, it is impossible to effectively monitor if compliance to the absolute maximum per year of \$14,000 is respected without full access to the information pertaining to party financing activities by the designated third party. What makes the situation even more complicated is that there are no provisions in place that provide the designated third party the means to coerce offenders. When it comes to practice, the assessment has revealed that there was no actual financial oversight taking place (Ibid.).

Another indicator that scored low was the indicator related to the integrity of the political party. Some information was required from all political parties in order to understand to what extent, in practice, political parties were effectively democratically governed. However, only one political party responded and thus there was no sufficient information for further assessment (Ibid.).

Finally, the assessment of the role specific indicators revealed that currently the political parties of Curaçao are inept at representing the interests of all in the political sphere. Also, although generally issues have been raised



by political parties regarding corruption and the importance of public accountability, no effort has been put toward the actual implementation of measures to combat corruption and improve public accountability (Ibid.).

Legislature

The legislature was amongst the other institutions that was also assessed in the NIS. Transparency International gave this institution an overall score of 46 (weak), with almost all indicators across all of the dimensions scoring low. The highest scoring indicators were **resources (law)** and **independence (law)** (Ibid.).

First of all, with regard to the resources (practice) indicator, the assessment revealed that while legally there exist many provisions that provide the legislature with sufficient financial, human and infrastructural resources for it to function properly. However, it was found that while in practice, the legislature is not able to receive enough funds for it to carry out its duties. There are some concerns that pertain to the adequacy of the necessary administrative support required to enable members to perform their duties (Ibid.).

Secondly, the assessment of the independence indicator showed that even though by law the parliament is deemed to be independent and free, in practice, this was not the case. It was observed that while parliament members did function independently in some circumstances, the parliament's independence seemed to be undermined by the party discipline (Ibid.).

Thirdly, with regard to transparency, the NIS revealed that while there are some provisions that require the parliament to ensure transparency of their activities (e.g. announce and report public sessions in a timely manner), no clear legal restriction were identified with respect to conventions behind closed doors. That parliament has the power to restrict any individual from entering its building is testament to this. It was also stressed that there were no provisions that require the parliament to publish parliamentary documents. Despite these gaps in providing transparency, it was found that the parliament complied to majority of the provisions. However, the obtainability of documents related to parliamentary activities and decision-making processes remained a challenge. Nonetheless, some improvements were observed with regard of accessibility of documents through the (then) new government website (Ibid.).

Further findings of the NIS assessment with regard to accountability showed that there are some provisions in place to ensure that parliament members are held accountable for their activities. However, according to the legal framework, the legislature is only to hold itself accountable to the kingdom. When it comes to the public, the legal framework does not require the legislature to hold themselves accountable to the public. Furthermore, the legal framework also gives limited immunity from prosecution to parliament members in the name of personal privacy and security. Similarly, it was revealed in the NIS assessment that next to almost never being held accountable for their actions, parliament members are also seldom held accountable by those that have the power to do so. Furthermore, according to the assessment, the public is also held in the dark when it comes to the actions of the parliament (Ibid.).

The NIS assessment also revealed, pertaining to the integrity of the legislature, that while the legal framework does contain some provisions that ensure the integrity of members of parliament (e.g. constitutional safeguards against conflicting public service positions and penalties against bribery), none were in place that enable the



regulation of gifts, assets, additional functions and lobbying contacts. Interestingly, some of these provisions were in practice prior to the constitutional change that took place in 2010. Unsurprisingly/As was foreseeable, the assessment concluded that the integrity of the legislature could not be ensured as a consequence of the lack of the above-mentioned provisions as well as the lack of transparency (Ibid.).

Finally, with regard to the indicators pertaining to the role dimension, the assessment showed that while the legislature (parliament), through the constitution, has the power and the responsibility to hold the activities of the executive (e.g. ministers) accountable for their actions. The parliament, according to the assessment: “does not pay a significant amount of attention to the Executive’s implementation of and compliance with legislation”. The Executive is not often held accountable for not complying with requirements to regularly inform Parliament on the execution of laws, or for not adequately implementing legislative provisions. As a result, as is illustrated in several chapters in this assessment, many provisions look good on paper but are not adequately implemented or complied with in practice. Also, on a more positive note, the assessment did reveal that the legislature has since the change of the constitutional relationship with the Kingdom of the Netherlands, adopted various regulations which aim to fight corruption in the country and promote good governance. However, it has not prioritized the monitoring of the implementation and compliance of said provisions (Ibid.).

Executive

The last institution that, according the NIS assessment, received a weak score was the executive (e.g. ministers and the governor). Overall, the score that was given to the executive was 43 (weak); With the majority of the indicators scoring either moderate or weak e.g. **resources (practice), Independence (practice), transparency (law and practice), accountability (practice), integrity (law and practice)** all of the **role specific indicators** (Ibid.).

First of all, it was shown in the NIS assessment that while in practice, executive did receive enough funds to perform its duties effectively, there remained a potential weakness in the selection of competent ministers. During interviews in the assessment, it was suggested that regarding the selection of ministers:

“sometimes loyalty to political leadership is considered more relevant than the ability to effectively carry out one’s duties”

Secondly, looking at the assessment of the independence of the executive, it was revealed that while the legal framework contains enough provisions which allows the executive to function in an independent manner, there is still the risk of executives to be influenced by lobbyist, since there are no provisions in place with regard to safeguarding the institution's independence in the wake of influence by lobbyists. In fact, the assessment showed that the decisions of the executive seemed in many cases to be influenced by political parties (Ibid.).

Furthermore, it was also noted that executives were in many instances approached by lobbyists. Nonetheless, it could not be determined to what extent this latter type of influence had an effect on the independence of the executive (Ibid.).

Thirdly, the assessment showed that when it comes to transparency, rules do exist in the legal framework that aim to safeguard transparency with regard to the government budgeting. However, the provisions which aim to ensure government transparency with regard to administrative decisions and all (non-confidential) documents



seemed to be less adequate. It was further established by this assessment that the legal framework hardly contained any transparency provisions concerning the meetings of the council of ministers. Moreover, it was found that despite the publishing of new laws and regular addresses to the press by the executive, it hardly provides the public with information about its activities. As a consequence, ambiguity ensues with regard to transparency pertaining to what legislation and/or power applies to whom and also how that power is used. Furthermore, the executive is not transparent when it comes to the information pertaining to its implementation of ordinances related to public access to government information (Ibid.).

The assessment further revealed that while there are many provisions in place which aim at holding the executive accountable for their actions, these provisions are not effective, since the parliament rarely questions the activities of the ministers (Ibid.).

Also, the NIS assessment showed that there were provisions in place that aimed at ensuring the integrity of the ministers. However, it seemed that little to no monitoring of compliance to these provisions was taking place. Furthermore, it was also brought to light that there were insufficient provisions in place for the protection of whistleblowers. Moreover, it was revealed that in practice, that there were still some vulnerabilities in the process outlined within the provisions for maintaining the integrity of members of the executive. Also, the effectiveness of the above-mentioned provisions could not be assessed, since there was a lack of effective oversight (Ibid.).

Finally, with regard to the role specific indicators, it was revealed that the executive structurally lacked a commitment to the create a well-developed public sector, although it was noted that a small amount of development had been observed/identified. Also, the assessment showed that some action was taken by the executive to strengthen the legal framework for the provision of good governance and the fight against corruption. It was noted however that some of the legislation still remained to be enacted and some already enacted provisions still remained to be implemented (Ibid.).

From the drawbacks touched on in the previous paragraphs, it can be concluded that the selection of potential monitors in Curaçao could be in some cases be categorized as nothing but smoke and mirrors. In such cases, this process is a great example of a centralized decision-making process which gets masked as a delegative one to the masses. This process could be extremely non-transparent and as such, may sometimes cause feuds or distrust amongst members of political parties. By the same token, many negotiations may occur behind scenes (Ibid.).

Hereby three additional examples of what could take place due to these political drawbacks are presented:

1. Influential groups might practice opportunism and lobby during times were there are no elections, fund parties for their elections and then profit when the time is appropriate and certain people have been appointed as ministers.
2. This concept of allegiances being formed between figures of political parties and third parties breeds a loyalty that goes beyond monetary benefits. Out of these relationships, these third parties often times are less critical towards certain political groups and may even start to give biased information in the case of media outlets.



3. An additional phenomenon that these allegiances are responsible for is the inactivity of certain political figures or rather the passive manner in which they conduct politics with even though this may not be their usual way of carrying themselves. This destroys any sense of moral coherence to the individual, any concepts or thought that go against the opinion of allies must be subdued before it causes any inconveniences.

When this is translated to our society, centralized decision-making may at times leave us with politicians with restricted hands, capabilities and a compromised sense of social justice which naturally comes at the expense of the majority (Ibid.).

Populism in Curaçao

While there is no direct single definition of populism, it does exist in different forms in different parts of the globe. The general idea however is that, populism is a political and social method that is used to foster an us-versus-them mentality, causing divide within nations or between regions. To attain a better understanding of this occurrence, populism is better identified in politics as a political party agenda that highlights the importance of “the people”, usually referring to the less fortunate or marginalized of the society, and assigning the blame well off (e.g. the elites). This definition in turn created a non-populist belief termed as elitism. As argued by Mudde & Kaltwasser: Elitists share the populist binary division but reverse the associations (Mudde & Rovira Kaltwasser, 2017). Where populists regard the elites as bad and the common people as good, elitists view “the people” as being vulgar, immoral, and dangerous and “the elites” as being morally, culturally, and intellectually superior (Mudde, 2004). Elitists want politics to be largely or entirely an elite affair. Populism can also be experienced in Curaçao as described as follows:

“In Curaçao the conditions for populism to develop are present. The elites are under heavy pressure. The great majority is poor and institutions are heavily criticized. The media in Curaçao are pervasive and invasive. Populism is growing. But the practical answers that populists provide appear to perpetuate the weaknesses of the original system, or reinforce them, while only the occupants of various positions change. This appears to be a feature of populism all over the world.”

This case study further highlights the various factors that contribute towards populism in Curaçao. These include a preference for direct democracy, anti-colonialism, ethnicity, socialism, religion, patronage and nepotism.

Political crisis

On October 10, 2010, the Netherlands Antilles was dissolved and Curaçao became an autonomous country within the Kingdom of the Netherlands. The powers and officials of the government of the Netherlands Antilles and the island territory of Curaçao were transferred to the new country Curaçao. On the same date, a new prime minister was appointed. Due to this occurrence, the screening of the Curaçao candidate ministers was not fully completed by the Netherlands Antilles Security Service (VNA) and instead it was decided that the investigations would be completed by the new Curaçao Security Service (VDC) after the Netherlands Antilles was dismantled and the candidate ministers were in their respective posts (Rosenmöller, Maas, & Hillebrink, 2011).



However, this investigation continued to stagnate; a stagnation which on the 27th of October 2010, drove the governor of Curaçao to exert more pressure on the Curaçao Security Service (VDC) to complete the screening investigation of the candidate ministers. The results of the full screening report were handed over to the governor of Curaçao on the 23rd of November 2010 (Ibid.).

The political crisis was sparked in Curaçao after a trusted memorandum of the results of this screening report which dated back to 27 October 2010 was leaked to the media in October 2011. This was subsequently published by various Curaçao and Dutch press companies (Aworaki, 2011; NOS, 2011). This trusted memorandum explains that there are so-called "security risks" attached to the appointment of some ministers of the Curaçao government (NOS, 2018). The respected positions of these ministers at that time were:

- Prime Minister of Curaçao of that time and Minister of General Affairs and Foreign Relations
- Minister for Economic Affairs and Development
- Minister for Health, Environment and Nature
- Minister for Transport and Spatial Development
- Minister for Education, Science, Culture and Sport

The security risks mentioned within the leaked memorandum were:

- Unknown domestic and foreign lenders
- Corruption
- Money laundering
- Interruption of the screening
- Association with organized crime
- Embezzlement
- Conflict of interest
- Bribery

This political crisis has led to the imprisonment of the former Prime Minister of Curaçao. In addition, the events have also led to further decline of the integrity of the governmental institutions of Curaçao. These events which have occurred are historical cases where centralized decision making power has been misused for the benefit of a few to the costs of many.

2.7.2 Socio-economic sphere

Like many other economies in the western hemisphere, the root socio-economic orientation of Curaçao's economy is based upon competition, self-interest, hierarchy, and ultimately, a sociology formed around scarcity, fear, and in-group/out-group bias (Joseph, 2017). This, in essence, is what has contributed to the socio-economic state Curaçao finds itself currently.



Nepotisms effect on FDI (Foreign Direct Investment)

Due to the small population of Curaçao there is no critical mass or economy of scale. The small-scale forges relationships that are extremely close, so that objectivity in decision-making is at a minimum. This has resulted in an increase in the tendency towards nepotism and patronage (Rosenmöller et al., 2011). Nepotism is the way politicians, companies and heads of public organizations structure their ‘networks’ (Sinica, 2012). This method of organizing in which loyalty is preferred over competence hinders development and therefore, innovation (Curaçao Chronichles, 2013). Another effect of this patronage is how it affects foreign direct investment – FDI. FDI is an investment made by an entity in a country with the intent of controlling ownership in a business or project in another country. The adaptation of loyalty over competence influences how individuals recruit their new staff or members, which most times would also be friends and family (Curaçao Chronichles, 2014). In that way, the investment needed for a new project would eventually come from within the exchange network instead of from the outside, albeit internationally. If the opposite was done, competence over loyalty, this would counter the urge to keep selecting individuals from the same network and when it comes to investment search, individuals would be more open to the idea of looking abroad. This, in turn, would bring more foreign influence and therefore, even more investors that are also looking to invest in other projects and eventually, bring development and innovation.

Shortcomings of the state

The state or more commonly known as the government can be simply described as an organization or a group of individuals who control and make decisions for a country. The state has various roles but a core purpose of the state from a socio-economic standpoint is to create protection and stability in commerce. This role is particularly important since this protection of and stability in commerce is not guaranteed by just relying on the notion of the “invisible hand” as proposed by Adam Smith (the father of modern-day economics) in two of his books *Wealth of Nations* and *The Theory of Moral Sentiments*. This argument was strengthened by Nobel prize-winning economist Joseph E. Stiglitz who famously stated (J. Stiglitz, 2006; J. E. Stiglitz, 2003):

“the reason that the invisible hand often seems invisible is that it is often not there.”

Furthermore, the state can also be used to secure competitive advantage through a select breed of large business powers. For example, the government of Curaçao is the partial owner of the energy production company Aqualectra. Now, because the company is state owned certain laws and regulations (also known as a government granted monopoly) can be circumvented which puts the company at a competitive advantage over any other organization that wants to produce energy (Aqualectra, 2019). These business powers can circumscribe government behavior overall, even though there is the notion that the government exists as an extension of the general public.

In the case of Curaçao, the government can be described as a democratic parliamentary state with a multi-party system. Elections are held every four years for a new government in which the population votes for a new parliament. There is, like in any organization, a built-in lack of trust in the state partly due to nepotism as mentioned before in combination with the lack of transparency within the entities that are supposed to audit the state (see 2.7.1), which can incentivize corruption (Transparency International, 2011). Although, corruption can be experienced in various areas of a society, when it comes to corruption within the state, this can directly



affect the socio-economic state of any country by breeding more corruption in other areas of the country. This scenario was highlighted in a simulation evolutionary game theory mentioned in the report “Corruption breeds Corruption” (Ghatak & Iyengar, 2014).

A similar scenario was experienced in Curaçao recently where there was an indictment of a subject who was charged with corruption, money laundering and paper fraud to support his goals (NU.nl, 2017). This is an example of what can go wrong when the decision-making power of an entity is centralized and therefore, requires more checks and balances with regards to ensuring transparency. If those checks and balances are not sufficient (as is the case of many governmental institutions in Curaçao, see previous section), subjects are free to go as they will within that entity and commit acts that do not benefit the community around them but merely themselves with little accountability or subjection to scrutiny by the public.

Challenges of socio-economic coverage

One of the challenges Curaçao faces is to first identify the corners of corruption and tackling it. However, these lie deep and due to the size of the island, if a journalist, for example, was to investigate and expose corruption, they could be faced with many obstacles. The first being the threat of physical violence. According to the Curaçao Chronicle, experts suggest that journalists on the island often practice the act of ‘self-censorship’ to diminish the risk of playing with fire (Curaçao Chronicle, 2013). This is also the reason why investigative journalism is something not commonly found in Curaçao. Furthermore, the political/legal environment on Curaçao is so intertwined by social connections, that if a ‘whistleblower’ were to emerge, the chances that he or she can be protected are slim. Also, it does not help that there are no legal provisions in place that allow for the protection of whistleblowers.

2.7.3 Environmental sphere

Curaçao as SIDS

SIDS are groupings of small developing island economies who share very specific socio-economic and environmental barriers when compared to the surrounding neighboring developed countries (Mohee et al., 2015). To date, there are 52 internationally recognized SIDS grouped together in three regions based on their geographical location, these include Caribbean, Pacific and AIMS (Atlantic, Indian Ocean, Mediterranean and South China) SIDS.

In Curaçao SELIKOR is the main municipal entity responsible for solid waste management. They are tasked with street and drain cleaning, waste collection, Transportation and unloading the collected waste on the designated landfill site (Mal Pais). SELIKOR is predominantly comprised of field workers with a few officers for planning and administration at their headquarters. In addition to waste collection, they are also tasked with finding innovative alternatives to the ever-increasing landfill.

Landfilling is Curaçao’s primary means of waste disposal because of the ability to accommodate all waste types and compositions coupled with a simple and cost-effective implementation (Seng, Hirayama, Katayama-Hirayama, Ochiai, & Kaneko, 2013). Malpais is a “first generation landfill” meaning that it does not apply modern



sanitary landfill protocols such as capping, layering. This leaves gas, liquid or solid waste dangerously exposed to the environment.

With increasing pollution comes increases in environmental degradation and public health concerns. To combat this socio-environmental stressor, steps should be initiated towards introducing national waste minimization/recycling strategies. These strategies should aim to reduce the highest fraction of waste being landfilled. According to (2E Afvalstof Plan Achtergrond Informatie 11-35) between 50% - 80% of Curaçao's MSW is comprised of organic-biodegradable waste. Its therefore imperative that solutions encompassing waste reduction strategies revolve around minimization of the organic waste. Moreover, socio-economic constraints of Curaçao should also be factored when formulating this strategy. Given all these conditions, both anaerobic digestion (AD) and organic composting (OC) are viable strategic pillars on which a national organic waste reduction program can be created.

Waste minimization/recycling strategies

Less organic waste at Malpais would not only prolong its lifespan, but also reduce the environmental impact the disposal site has on the surrounding area. Organics are largely responsible for the polluting leachate, methane, odor problems and public health concerns. The implementation of anaerobic digestion or composting can be construed as a plausible alternative, improving both the environment as well as the livelihood of local inhabitants. Nevertheless, these alternatives are dependent on factors such as market demand for the end products (gas, digestate or compost), as well as the technical and organizational set-up of the individual facilities. These technologies are cost-effective, scalable and have proven successes in other Caribbean SIDS (Edghill, 2012; Meyer-Kohlstock, Hädrich, Bidlingmaier, & Kraft, 2013). The reuse of the organic waste fraction through OC or AD on Curaçao is a new concept. Composting and AD, unlike gasification and combustion are very energy efficient.

Organizational capacity building correlates with the development of efficient and effective operations of institutions and organizations (Friedrichs, Logan, & Malik, 2002). Institutional capacity is generally built by strengthening individual organizations, providing technical and resource support, ensuring integrated planning and effective decision-making processes within and across multiple institutions (Mosler, Drescher, Zurbrügg, Rodríguez, & Miranda, 2006). In the context of waste (environmental) management, this implies that clear and enabling legislation, municipal and public commitments are imperative to the successful adoption of either AD or OC.

In a stakeholder analysis, Gilbert, L. concluded; that the local population has a relatively high waste consciousness with the majority of the stakeholders committing to recycling if the option was more readily available (Gilbert, 2016). In contrast, SELKOR is unable to undertake a task of this magnitude. Therefore, many NGO's have taken it upon themselves to educate the rest of the community in response to SELKOR's lack of involvement. However, these bottom-up initiatives often operate with little to no support or cooperation with each other. There seems to be a gap in organizational capacity of NGO's and citizens. This gap, coupled with the above-mentioned bureaucratic bottleneck, leads to a weak institutional environment and renders the introduction of suitable/sustainable waste management ineffective. Curaçao's waste management issue –



contrary to popular belief – does not revolve around financial constraints. Rather, innovation is being hampered by the lack of cross-sectoral cooperation.



3 Methodology

3.1 Theory analysis

For this research, a couple of qualitative methods were used to analyze the relevant theory as found in the theoretical background. The following method was applied in this report: Document analysis. Document analysis is a form of qualitative research in which documents are interpreted to give voice and meaning around the research topic. Hereby relevant texts, reports and books were gathered. These were assessed for their authenticity, explored for their agenda/biases. The choice of document analysis as the research method for this research was firstly, to provide data on the context within which research participants operate—a case of text providing context, if one might turn a phrase. Bearing witness to past events, documents provide background information as well as historical insight. Secondly, this choice was informed by the fact that the information contained in documents can suggest some questions that need to be asked and situations that need to be observed as part of the research. Thirdly, documents provide supplementary research data. Information and insights derived from documents can be valuable additions to a knowledge base. Additionally, documents provide a means of tracking change and development. Lastly, documents can be analyzed as a way to verify findings or corroborate evidence from other sources (Bowen, 2009).

In sum, documents provide background and context, additional questions to be asked, supplementary data, a means of tracking change and development, and verification of findings from other data sources. Moreover, documents may be the most effective means of gathering data when events can no longer be observed or when informants have forgotten the details.

Other sources that were used to answer and argue the research questions were books and academic articles. The shortcomings of decentralized decision-making were identified mostly through inference. The assumptions that were taken were based on the general idea that the institutions (organizations) mentioned in the theory were organizations that have a central decision-making organizational structure (i.e. these organization have a hierarchical organizational model) and thus had management procedures that were based top-down decision-making approaches.

From the theory pertaining to each of the spheres analyzed (environmental, socio-economic, socio-political), some shortcomings were identified. The shortcomings were then further broken down as to how and why these shortcomings existed. These explanations are the result of logical inferences made from the analysis of the evidences from the selected theory.

3.2 Theoretical DAO concept

After the shortcomings were identified, a visual concept was created based on the theory pertaining to DAOs. A conceptual design is the early phase of a design process in which the broad outlines of function and form of something are articulated and visually represented. Each shortcoming was analyzed in the context of how certain characteristics (e.g. the blockchain, governance etc.) pertaining to DAOs could possibly mitigate the shortcomings (i.e. act as a possible solution). This part of the analysis was based on the elements of the different



DAO frameworks that were studied. The frameworks provide a range and variety of organizational possibilities that can be utilized to extrapolate the probable functions that can be embedded in the DAO concept.

The following smart contract frameworks were utilized for constructing the conceptual design e.g. Colony, Alchemy (Daostack) and Aragon. First, Colony is a new framework and is still in development. However, it has a clear structure that are explained very well and thus can be used to construct the conceptual design. Daostack's Arc on the other hand, has been in development for quite some time already and as such, has already deployed a couple of versions of DAOs built on this framework. (e.g. the Genesis DAO, dxDAO, PolkaDAO Kyber DAO, dOrg DAO). Since Daostack's Arc has been in development for a considerable time already and contains sufficient components that can be utilized for the construction of the design. Finally, Aragon was also selected as a possible framework for the creation of the concept. The reason for this is that the framework allows for a lot of freedom when creating a DAO. This is primarily because the framework allows one to set particular roles to each agent or group of agents in the DAO in the context of the delegation of power, does providing the community with a wider range of tools that can be utilized for improving the organizational capacity of the community. Furthermore, the company behind the framework has created an interface that makes it easy to launch DAOs.



4 Results

4.1 Theory analysis

From the analysis of the information pertaining to the different spheres that were studied, shortcomings were identified (see Table 4).

Table 4 Identified shortcomings in the three spheres

Sphere	Shortcomings	Description
SOCIO POLITICAL	Accountability	Accountability-related issues in governmental institutions (e.g. judiciary, ombudsman, electoral management body, political parties, legislature and executive)
	Transparency	Transparency-related issues in governmental institutions (e.g. ombudsman, electoral management body, political parties, legislature and executive)
	Corruption	Corruption-related issues in governmental institutions (e.g. judiciary, political parties and legislature)
	Social Misrepresentation	Possibility of unfair and unequal representation of all people in the social structure.
	Populism	Conditions for the rise of populism are ever present in Curaçao
SOCIO ECONOMIC	Nepotism	The adaptation of loyalty over competence
	Corruption	Corruption as a result of power imbalance with regard to relationship between the state and the market.
	Accountability	Reduced accountability of “bad players” in the economic system.
ENVIRONMENTAL	Cooperation	Lack of cross-sectoral cooperation, which leads to the misuse and/or the failure to adopt necessary technology for solving the waste management problem in Curaçao



4.1.1 Socio-political sphere

Out of the analysis of the theory, four shortcomings were identified in the social political sphere namely: **accountability, transparency, corruption, social misrepresentation** and **populism**.

Accountability

It was found that many institutions in government; particularly, the institutions that are part of the national integrity system were experiencing accountability issues. It seemed that generally, while in most cases there were provisions with respect to holding the actors or actions of these actors within the institutions accountable. In practice, these provisions were not always sufficient or in some cases, rarely applied. Furthermore, in one case (e.g. political parties), the legal framework contained loopholes that would allow the players within the institutions to avoid being held accountable. What was also observed in some cases was the lack of independent organizations that are tasked with monitoring compliance to these provisions.

Transparency

Another shortcoming that was identified was lack of transparency. While most of the institutions of the national integrity system of Curaçao do have laws pertaining to maintaining good transparency practice, in practice, these provisions fail to be implemented efficiently and effectively. A situation is created in which institutions and people within those institutions have poor transparency practices with regards to government activities, finances, laws and regulation and also fighting corruption.

Corruption

Corruptibility was amongst other shortcomings also identified in the assessment of the theory related to centralized decision-making structures in the socio-political sphere of Curaçao. There is a very real probability that these institutions or actors (e.g. national integrity system of Curaçao) within these institutions will behave in corrupt ways. This is the consequence of the state these institutions, in an organizational context, find themselves in. It appears that a number of institutions on the socio-political spectrum are experiencing limitations with regard to minimizing corruption. In most of the cases there is either an inefficient legal framework for example penalizing corruption, doing financial audits or monitoring activities. In other cases, there is an inefficient application of the legal framework by the actors within the institutions themselves. This creates an environment where corruption can proliferate or persist.

Social misrepresentation

Another shortcoming that was found was that of social misrepresentation. This was found to be a (direct) result of the power that companies, or organizations are capable of wielding in the political sphere as a result of possible financial influence that is disproportionate to that of the public. As a consequence, the fair and equal representation of the interests of all of the people in the social structure cannot be assured.



Populism

One other shortcoming that was identified was populism. With socio-economic conditions deteriorating due to the cynical centralized powers structures that have shaped in Curaçao over time, as well as a great majority considered under the poverty line, an ever-growing income gap between the upper class and middle-lower class, the oil refinery on the brink of shutting down and this event affecting the income of around 4.000 individuals, Curaçao's socio-economic situation could continue to deteriorate even further. Furthermore, the continuous lack of transparency, accountability and distrust staining Curaçao's national integrity system (see 2.7.1) could further set the stage for populism to rise in Curaçao. It seems that growing inequality will only worsen the situation.

4.1.2 Socio-economic sphere

Out of the analysis of the theory three shortcomings were identified in the social economic sphere namely: **Nepotism, Accountability and Corruption.**

Nepotism

Nepotism or patronage was highlighted as a shortcoming due to the adaptation of loyalty over competence in Curaçao. This influences how high and low ranked individuals within the economy structure their networks by recruiting their new staff or members, which most times would be friends and family. This can be experienced due to the lack of foreign investment in local projects to bring more development and innovation opportunities e.g. crowdfunding initiatives.

Accountability

Accountability is a shortcoming due to the lack of investigative journalism about cases where private and public organizations try to bend the economic rules for self gain. The political/legal environment on Curaçao is so intertwined by social connections that if a 'whistleblower' were to emerge, the chances that he or she can be protected are slim. Also, it does not help that there are no legal provisions in place that allow for the protection of whistleblowers. This is the reason why journalists on the island often practice the act of 'self-censorship' to diminish the risk of playing with fire. If public and private watch dogs (the public press & independent press) avoid documenting and investigating malicious economics players, then it is hard to hold heads of companies, politicians and institutions accountable.

Corruptibility

With the shortcomings of investigative journalism and other auditory private & public institutions (accountability & transparency) and the lack of trust in the state due to nepotism (mentioned in the socio-political section), there is a higher possibility for corruption to arise within the economy of Curaçao, as the participants within every economy are correlated. Corruption within the state (public) can breed corruption in the private sector and vice versa. The interconnected nature of economics can cause the lack of integrity and complexity when it comes to internally changing Curaçao's (and many other countries) social shortcomings within their economic framework, to a more resilient, inclusive and progressive economy.



4.1.3 Environmental Sphere

Out of the analysis of the theory, one key shortcoming was identified in the environmental sphere which was **Cooperation**.

The shortcoming experienced in the environmental sphere of Curaçao (waste management issue) – contrary to popular belief – does not revolve around financial constraints. Rather, innovation is being hampered by the lack of cross-sectoral cooperation. As mentioned before, the local population has a relatively high waste consciousness, with the majority of the stakeholders committing to recycling if the option was more readily available. In contrast, SELIKOR is unable to undertake a task of this magnitude. Therefore, many NGO's have taken it upon themselves to educate the rest of the community in response to SELIKOR's lack of involvement. However, these bottom-up initiatives often operate with little to no support or cooperation with each other.

4.1.4 Connecting the dots

It cannot be stated with certainty (because of the lack of quantitative data) that centralized decision making is the direct cause for the shortcomings that have been identified in the analysis of the theory. However, logical inferences can be made when perceiving the centralized decision-making in the social context.

On a first note, these shortcomings seem to be recurring issues in modern societies that organize their prosperity through centralized decision-making. Two recurring issues that can definitely be attributed to centralized (hierarchical) organizing are the issues of transparency and corruption. Often enough, the lack of the former leads to an increase in occurrence of the latter. For example, a 2014 report on institutional transparency conducted by Transparency International concluded that: "90 [out of 101] companies on the world (largest companies traded on the stock market) fail to reveal any information about tax payments in foreign countries (Kowalczyk-Hoyer, Akhmadullina, Begrman, Kwok, & Sakai, 2012). In another example, the same organization also published in 2018, the corruption perceptions index. The index measured the perceived corruption in the public sector in 180 countries across the globe, with scoring ranged between 0 (highly corrupt) - 100 (very clean). It was found that around two thirds of the countries scored below 50 with the average being 43. Transparency International further concluded that: "the vast majority of the countries assessed have made little to no progress" (Transparency International, 2018). The data does support the assumption that when institutions are transparent about their practices, they can be held accountable by the public. Public scrutiny can lead to the exposure of corruption and the possible reduction thereof through organizational reform or legislation. Furthermore, a high level of transparency can fend off corruption altogether since no player wants to face public shame and legal action against their unethical behaviors. Contrarily, a low level of transparency can lead to an increase of corruption since bad behavior cannot be reported as a result of lack of information pertaining to, for example, financial practices of players.

On a second note, it can be argued that centralized hierarchical models of institutional organizing can lead to a higher probability for the occurrence of the shortcomings identified in the previous section. This mainly attributable to the bureaucratic complexity that is created as a result of organizational capacity (e.g. governance, law and regulation) that is required ensure the integrity of the institutions. For example, consider the



shortcomings in transparency, accountability and corruption of the national integrity systems of Curaçao. In order to secure the integrity of the NIS as a whole, many third parties, watchers so to speak (e.g. legislature, executive and judiciary) are required to ensure that public sector and the non-governmental actors behave in an ethical manner (that being: to not harm the public, cheat the public, cause damage to the environment etc.). Moreover, the watchers themselves have to watch each other so that unethical behavior is prevented. The problem occurs when the watchers themselves are susceptible to the shortcomings mentioned above (as was revealed in analysis of the theory). Also, it can be argued that these shortcomings are inherent in centralized hierarchical organization as a result of the power imbalance/distance that is created amongst/between the different ranks of people within these types of organizations.

On a final note, it seems that the occurrence of these shortcomings is not necessarily the result of unethical behavior from bad players. In essence, these shortcomings are of a structural nature, and while centralized organization is not the definite cause, it does create an environment in which the probability for the occurrence of these shortcomings can persist. It can be justifiably assumed that the power imbalances, organizational complexity and distance between workers and bosses or between the governors and the governed, allow for the occurrence/proliferation of issues in the related to accountability, corruption, transparency and cooperation. These issues manifest as systemic symptoms like ex. populism or as problems in FDI.

4.2 Theoretical DAO concept

The shortcomings were used as a guiding hand for constructing a theoretical concept (see Figure 4) with the aim of addressing the shortcomings identified in the previous sections. The figure is a conceptual construction of a DAO that is comprised of different concepts sourced from blockchain organizing. It is comprised of three layers namely; a **Distributed trust network**, an **Application layer** and a **Decentralized Autonomous Organization (DAO)** – which is represented by an application that rest on the bottom two layers.

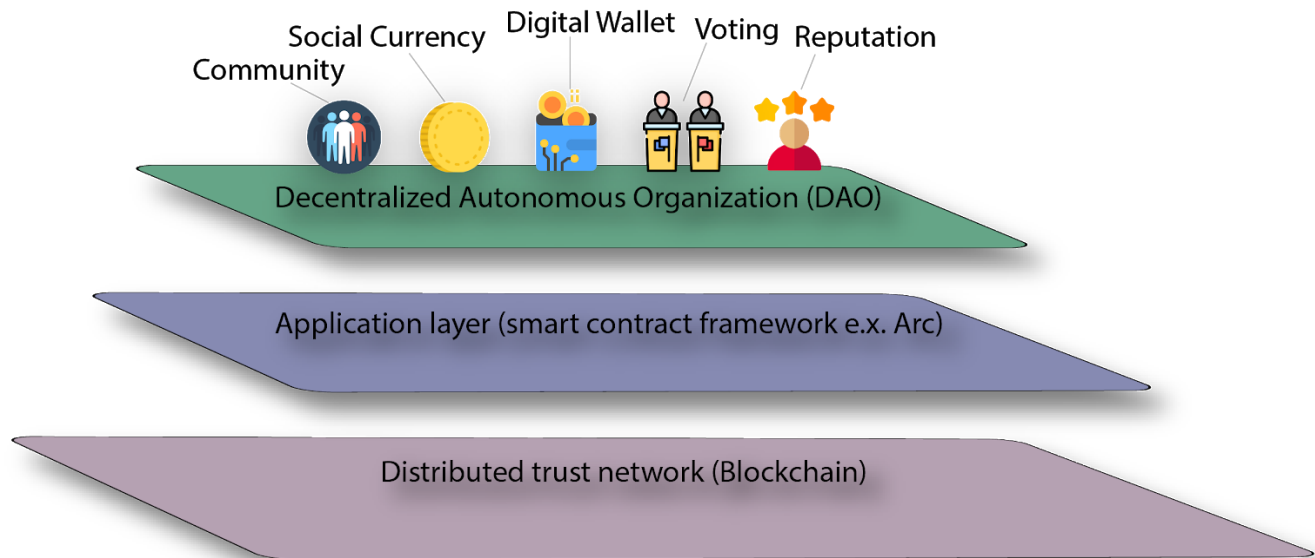


Figure 4 DAO conceptual design

4.2.1 Distributed trust network (blockchain)

The bottom layer (first layer) of the DAO is what is called a distributed trust network. This layer represents a distributed network of computers tasked with the maintenance of a cryptographically secured and immutable blockchain - a chain of cryptographically secured and validated transactions stored as blocks of data (i.e. a so-called ledger), in which each successive block refers to the previous blocks with a cryptographic digest (hash). A blockchain created by a DTN usually provides information (e.g. transactions) that, by design, is immutable and subject to provenance; Thus, providing trust based on the mechanisms that constitute how the network achieves consensus (e.g. consensus mechanism) on the state of the blockchain (ledger). These qualities of the blockchain offer multiple potential solutions for the shortcomings identified in the previous sections.

One of the first shortcomings that can be addressed with the use of a blockchain is **transparency**. Since all of the transactions that have taken place within the network are recorded on the blockchain and as long as the network is an open network (i.e. the blockchain (ledger) on which the transactions are recorded are accessible to everyone within the network), transparency is guaranteed. Generally, by design there is no need for legislations or provisions to ensure transparency with regard to the transactions that have taken place on the network. Essentially, these provisions are programmatically enforced through computer protocol instead of a third party.

Another shortcoming that can be addressed by utilizing a blockchain is **accountability**. Essentially, because each transaction is retraceable as a result of the digital signatures required to send a transaction, actors in the network can be held accountable for their behavior because anyone in the network can see who the sender and receiver of a transaction is. However, the blockchain on its own is not enough to ensure accountability of actors within the network, other components are required e.g. applications that allow for the monitoring of actors. Contrary to an organization or network of organizations, provisions or legislations for safeguarding accountability are programmatically encoded and enforced by a computer protocol instead of third parties.



Next, **corruption** is also a shortcoming that can be addressed by the blockchain layer. However, this can be most efficiently achieved with other components on top of the blockchain that allow for the monitoring of behavior and the punishment of “bad behaviors” (more about this in the following sections). At the most fundamental level, the blockchain can allow for minimized corruption because of the immutability and transparency that are inherent in the system. Again, no third parties are required to provide corruption security, through a legal framework.

Finally, the blockchain layer can also address the problem of cooperation between institutions as was identified in the environmental sphere. Fundamentally, a blockchain is governed by one “ontology” i.e. one set of rules, one data model or one IT infrastructure that is applied to all nodes in the network, essentially creating an atmosphere in which cooperation can be fostered because of the increase in coordinative capacity allowed by the blockchain. For example, nodes (which can be represented by humans) in the network do not need to use different payment systems, databases, HRM systems to cooperate with each other.

4.2.2 Application Layer (smart contract layer)

The application layer is represented by the smart contract framework that underlies the different components (e.g. social currency, voting, digital wallet etc.) that comprise the DAO. Components used will differ depending on the type of DAO that one wants to create. As is explained in section 2.6, these frameworks are basically libraries of prewritten chunks of code (e.g. smart contracts) that can be utilized to build the DAO. The kind of shortcoming that is to be addressed depends on the type of DAO that is created. It is important to note that without the application layer it would take a lot of time and effort to create a DAO from scratch. Thus, the application is imperative to addressing the remainder of the shortcomings that are identified.

4.2.3 Decentralized Autonomous Organization (DAO)

The upper layer -which exists due to the lower layers - is what can be described as the Decentralized Autonomous Organization (DAO). In this concept, it is comprised of four major components: **community**, **social currency**, **digital wallet**, **voting** and a **reputation system**.

First of all, there is the community: this is the most integral part of the DAO. Each community member acts as an agent in the DAO. Without the community actively engaging with each other through the use of the other components that represent the DAO, it ceases to exist.

The second component of the DAO is a social currency. A social currency is any digital token or currency that is utilized in transactions between agents. These transactions do not always have to be strictly monetary, they can also represent collective decisions (consensus) achieved by all of the community members in the DAO.

Thirdly, there is a digital wallet. While each agent within the DAO has his/her own digital wallet of which its keys can be used for access to the DAO, in many cases there is also a digital wallet represented by smart a contract (or group of smart contracts) that can be completely autonomous (if programmed to be so). The wallet can represent the DAO itself and can act as the impartial agent (i.e. neutral third party) that completes actions (ex.



once consensus has been reached through voting, for example, on a particular action that needs to be taken by certain agents, release x amount of value to the parties tasked with the completion of the action).

One more component that is a part of the created concept is a voting mechanism. Combined with a social currency that for example represents voting power, the voting mechanism can be utilized by agents to reach consensus on decisions regarding the DAO itself or actions (ex. project proposals) that are to be completed by proposing agents.

Next, the design also constitutes a reputation system. The reputation system is primarily used for the provision of security with regard to accountability of the agents that comprise the DAO. A reputation system is imperative in this design because it forces agents to behave according to the programmed rules of the DAO and the principles set by the DAO members. Because the rules are programmatically enforced, no use of external forces (e.g. impartial third parties) is required to secure the integrity of the DAO. For example, if a member does not deliver on his/her promises, this behavior can then be reflected in his reputation and does will affect the acceptance of future proposals made by the agent. However, for the DAO to maintain its impartiality (and thus integrity), the rules have to defined in an open manner with the input of all if the agents that are to be affected. If this is not the case, then the DAO could lose its impartiality and thus its integrity. The last thing desired is for the rules to reflect the interest of only a certain group of agents in the DAO.

Addressing the shortcomings

Similarly, to the blockchain layer, the DAO itself could theoretically tackle many of the shortcomings that were identified in the previous sections albeit in different contexts.

First of all, and perhaps with great importance, two of the major shortcomings that can be addressed with the use of a DAO are **accountability and corruption**. If all of the components as described by the concept are in place, this could by design lead to improved accountability of agents in the DAO. Bad behavior can be appropriately punished through the reputation system and thus agents that are not acting according to the rules set by the community can more effectively be held accountable. If actors know that they can be held accountable (as a result of transparency by design) and that this accountability can be enforced without them being able to influence the enforcers (since the enforcement is secured by design) then there is no gain for actors to misbehave, thus potentially leading to the reduction of corruption.

Secondly, **cooperation** can be appropriately improved through the use of a DAO. This is because all of the agents in the DAO can cooperate with each other without friction because of the homogeneity created when utilizing IT infrastructure and/or management procedure. Agents can easily transact with each other through a DAO in whatever form without the need of creating contractual agreements secured and enforced by third parties. These contractual agreements are programmatically represented by the components of the DAO (e.g. the voting system and the reputation system).

Thirdly, another shortcoming that could be addressed by the DAO is **social misrepresentation**. Generally social misrepresentation is the result of certain actors (usually the public or the workers) being able to influence the system through lobbying efforts or covert means which results in the creation of rules and laws that



disproportionately favor those that are most successfully able to lobby for their interests. In a DAO, this “representation imbalance” is mitigated because all the actors involved can act as active witnesses to the decisions (and adherence to those decisions) that are made collectively. In an open DAO, anyone can scrutinize the code that underlies, for example, the reputation system or the voting mechanisms that underlie the DAO. Furthermore, anyone can give their input on how these rules should be encoded, thus creating a more balanced form of governance. This is in stark contrast to conventional hierarchical organization structures in which actors can make decisions on behalf of all the agents behind closed door meetings.

Next, **populism** could also potentially be addressed should economic organizing be realized through DAOs. This is primarily because populism is a symptom of a system that has failed to cater to the needs of everyone and instead caters only to those that have the power to influence. If economic organizing was to be realized through DAOs, of course those that are programmed to represent (or work in favor of everyone) instead of just a small minority, in other words, DAOs that are not privately owned or to which its governance cannot be influenced by all actors that are involved, then the power would be more evenly distributed and thus the outcomes would reflect the needs of all instead of a minority. This would mean that populists would have less of an opportunity to appeal to the anger and frustration of those that feel misrepresented because people's needs would be met in a more equal manner (as a result of improved governance in economic organizing).

Finally, **nepotism** could also be addressed with the use of DAOs in economic organizing. This could be the consequence of collective decision making that is inherent in open DAOs. Agents would be chosen to do certain tasks if they are deemed to be qualified to complete those tasks by all the actors in the DAO. Furthermore, if the actors fail to complete tasks or if they provide incomplete work then they could be easily held accountable with the help of the different components that underlie the DAO as proposed in this concept.

Discussion

What needs to be firstly considered is that this research paper is merely an attempt at understanding the shortcomings of centralized decision-making in the context of economic organizing within one state alone. Therefore, it should be noted that there could be more shortcomings that were missed in the theoretical background since the research focused solely on certain sectors within Curaçao. In order to get a complete understanding of the faults of centralized decision-making, one needs to include the analysis of multiple sectors from a range of multiple countries.

Next, it should also be noted that with regard to the smart contracts framework available for constructing DAOs, there are more of these types of frameworks that were probably overlooked. Future research should include more of the frameworks that are present and dive more deeply in them. Also, many of these frameworks are fairly new and thus are bound to be faced with issues regarding the implementation and use of the DAO components that underlie these frameworks.

Also, the concept of DAOs is fairly new (about a decade) and thus there is not sufficient information available on the success rate of DAOs. In fact, it is even safe to assume that there are still a lot of issues that need to be



overcome regarding the successful function of a DAO. Furthermore, precaution should be taken with regard to which direction the development of DAOs should go. Clearly, there are justifiable concerns (see 2.5) that need to be considered when organizing DAOs which include: drawbacks related to the class relationships that will be affected by DAOs, the possibility of further capital concentration and thus, the concentration of power and the effects of automation. These concerns can only be fully addressed if the DAO is created and run in an open and inclusive manner, meaning a manner in which all the agents are actively involved in the creation and management of the DAO.

Finally, further research should be done in order to understand the full range of benefits and drawbacks pertaining to the use of DAOs. This could be achieved through the analysis of existing DAOs, defining the different types of DAOs and looking at the success rates of these DAOs. However, as was mentioned before the concept of DAO's is fairly new and thus more time will be required in order to find a sample broad and sufficient enough for analysis.

Conclusion

First of all, shortcomings were found through the analysis of outcomes with regard to centralized decision making in economic organizing Curaçao. The findings suggest that there are shortcomings in all of the spheres that were analyzed which include: transparency, accountability, cooperation and corruption. Furthermore, these shortcomings translate into the following social issues (symptoms): social misrepresentation, nepotism and populism.

From the analysis of the shortcomings and the possible solutions proposed by DAOs a conceptual design was created. The conceptual design comprised of three layers, the blockchain layer, the application layer and the DAO layer. Jointly, by design, these show potential solutions to the shortcomings related to transparency, accountability, corruption and the problems that are a consequence of those shortcomings e.g. social misrepresentation, nepotism and populism.



References

- Agre, P. E. (2009). Hierarchy and History in Simon's "Architecture of Complexity." *Journal of the Learning Sciences*, 12(3), 413–426. https://doi.org/10.1207/s15327809jls1203_4
- Aqualectra. (2019). Corporate organization - Aqualectra. Retrieved June 12, 2019, from <https://www.aqualectra.com/en/corporate-organization/>
- Aragon. (2018a). Aragon SF Presentation.pdf - Google Drive. Retrieved October 30, 2019, from <https://drive.google.com/file/d/1LVwdmaI941CI0eHA-09QAohbziwHhLYM/view>
- Aragon. (2018b). SFCDevs Presents: From the Eyes of an Eagle: The Technical Landscape of Aragon - YouTube. Retrieved October 30, 2019, from <https://www.youtube.com/watch?v=jFjWjkhLPZ8&feature=youtu.be>
- Aragon. (2019a). Aragon Frontend Walkthrough - Aragon One blog. Retrieved October 4, 2019, from <https://blog.aragon.one/aragon-frontend-walkthrough/>
- Aragon. (2019b). aragonOS intro · Aragon Developer Portal. Retrieved October 30, 2019, from <https://hack.aragon.org/docs/aragonos-intro>
- Aragon. (2019c). dao-templates/templates/company-board at master · aragon/dao-templates · GitHub. Retrieved October 30, 2019, from [https://github.com/aragon/dao-templates](https://github.com/aragon/dao-templates/tree/master/templates/company-board)
- Aragon. (2019d). GitHub - aragon/dao-templates: Monorepo for all Aragon made organization templates. Retrieved October 30, 2019, from <https://github.com/aragon/dao-templates>
- Aworaki. (2011). Schotte en vier ministers niet door screening. Retrieved from <https://web.archive.org/web/20141129084739/http://internationaljustice.rnw.nl/caribiana/article/scotte-en-vier-ministers-niet-door-screening>
- Blau, P. M. (1968). The Hierarchy of Authority in Organizations. *American Journal of Sociology*, 73(4), 453–467. <https://doi.org/10.1086/224506>
- Bowen, G. A. (2009). Document Analysis as a Qualitative Research Method. *Qualitative Research Journal*, 9(2), 27–40. <https://doi.org/10.3316/QRJ0902027>
- Brafman, O., & Beckstrom, R. A. (2006). The starfish and the spider : the unstoppable power of leaderless organizations. Portfolio. Retrieved from <https://duckduckgo.com/?q=The+starfish+and+the+spider&atb=v151-1&ia=shopping>
- Bril, W., & Bessems, P. (2016). Blockchain Organisieren.
- Bunderson, J. S., van der Vegt, G. S., Cantimur, Y., & Rink, F. (2016). Different Views of Hierarchy and Why They Matter: Hierarchy as Inequality or as Cascading Influence. *Academy of Management Journal*, 59(4), 1265–1289. <https://doi.org/10.5465/amj.2014.0601>
- Buterin, V. (2014). DAOs, DACs, DAs and More: An Incomplete Terminology Guide. Retrieved June 12, 2019, from <https://blog.ethereum.org/2014/05/06/daos-dacs-das-and-more-an-incomplete-terminology-guide/>
- Centraal Bureau voor de Statistiek. (2008). Berekening van de armoedegrens voor Curaçao.
- Chen, K., Stegorean, R., & Nistor, R. (2017). A Modern Management Approach in Internet Era. 11th International Management Conference, 918–927.
- Colony. (2019). Organizations, for the Internet | Colony. Retrieved October 4, 2019, from <https://colony.io/>
- Curaçao Chronichles. (2013). Marvelyne Wiels hires employees without application procedures - Curaçao Chronicle. Retrieved June 12, 2019, from <http://curacaochronicle.com/main/marvelyne-wiels-hires-employees-without-application-procedures/>
- Curaçao Chronichles. (2014). KKC-blog | Machtsstrijd Curaçao ontbrandt: Marvelyne Wiels: corruptie heeft een gezicht | Knipselkrant Curacao. Retrieved June 12, 2019, from <https://www.knipselkrant-curacao.com/kkc-blog-machtsstrijd-curacao-ontbrandt-marvelyne-wiels-corruptie-heeft-een-gezicht/>
- Curaçao Chronicle. (2013). Curaçao: small island, big challenges? - Curaçao Chronicle. Retrieved June 12,



- 2019, from <http://curacaochronicle.com/columns/curacao-small-island-big-challenges/>
- DAOstack. (2018). An Operating System for Collective Intelligence. Retrieved from <https://daostack.io/wp/DAOstack-White-Paper-en.pdf>
- Edghill, L. (2012). Report on the 6th Caribbean Environmental Forum St Kitts and Nevis May 21- 25, 2012. Retrieved from <http://businessbarbados.com/wp-content/uploads/2012/08/Report-on-the-6th-Caribbean-Environmental-Forum.pdf>
- Falkon, S. (2017). The Story of the DAO — Its History and Consequences. Retrieved June 12, 2019, from <https://medium.com/swlh/the-story-of-the-dao-its-history-and-consequences-71e6a8a551ee>
- Freund, A., & Stanko, D. (2018). The Wolf and the Caribou: Coexistence of Decentralized Economies and Competitive Markets. *Journal of Risk and Financial Management*, 11(2), 26. <https://doi.org/10.3390/jrfm11020026>
- Garrod, J. Z. (2016). The real world of the decentralized autonomous society. *TripleC*, 14(1), 62–77. <https://doi.org/10.31269/triplec.v14i1.692>
- Ghatak, A., & Iyengar, S. (2014). Corruption Breeds Corruption. *Studies in Microeconomics*, 2(1), 121–132. <https://doi.org/10.1177/2321022214522730>
- Gilbert, L. (2016). Waste to Energy Islands: Is converting waste to energy more feasible then traditional landfills for small islands, like Curacao? Wageningen Univerity.
- Hoogervorst, J. A. P., Koopman, P. L., & Van Der Flier, H. (2005). Total quality management: The need for an employee-centred, coherent approach. *TQM Magazine*, 17(1), 92–106. <https://doi.org/10.1108/09544780510573084>
- Hsieh, Y.-Y., & Vergne, J.-P. (2018). Bitcoin and the rise of decentralized autonomous organizations. *Journal of Organization Design*, 7(1), 14. <https://doi.org/10.1186/s41469-018-0038-1>
- Hutt, R. (2016). What are the 10 biggest global challenges? Retrieved June 12, 2019, from <https://www.weforum.org/agenda/2016/01/what-are-the-10-biggest-global-challenges/>
- International, T. (2011). Transparency International is the global civil society organisation leading the fight against corruption . Through more than 90 chapters worldwide and an international secretariat in Berlin , we raise awareness of the damaging effects of corruption and wo. Retrieved from www.transparency.org
- Joseph, P. (2017). The new human rights movement : reinventing the economy to end oppression.
- Kowalczyk-Hoyer, B., Akhmadullina, L. D., Begrman, T., Kwok, K. S. T., & Sakai, J. M. (2012). Transparency in Corporate Reporting: Assessing the World’s Largest Companies. <https://doi.org/978-3-943497-22-9>
- Matroos-Lasten, L. (2018). Achieving a higher and more inclusive growth path for Curacao (p. 17).
- McDowell, T., Agarwal, D., Miller, D., Okamoto, T., & Page, T. (2016). Organizational models: A network of teams | Deloitte Insights. Retrieved June 12, 2019, from <https://www2.deloitte.com/insights/us/en/focus/human-capital-trends/2016/organizational-models-network-of-teams.html#endnote-sup-14>
- Meyer-Kohlstock, D., Hädrich, G., Bidlingmaier, W., & Kraft, E. (2013). The value of composting in Germany – Economy, ecology, and legislation. *Waste Management*, 33(3), 536–539. <https://doi.org/10.1016/j.wasman.2012.08.020>
- Mohee, R., Mauthoor, S., Bundhoo, Z. M. A., Somaroo, G., Soobhany, N., & Gunasee, S. (2015). Current status of solid waste management in small island developing states: A review. *Waste Management*, 43, 539–549. <https://doi.org/10.1016/j.wasman.2015.06.012>
- Mosler, H. J., Drescher, S., Zurbrügg, C., Rodríguez, T. C., & Miranda, O. G. (2006). Formulating waste management strategies based on waste management practices of households in Santiago de Cuba, Cuba. *Habitat International*, 30(4), 849–862. <https://doi.org/10.1016/j.habitatint.2005.09.008>
- Mudde, C. (2004). The Populist Zeitgeist. *Government and Opposition*, 39(4), 541–563. <https://doi.org/10.1111/j.1477-7053.2004.00135.x>
- Mudde, C., & Rovira Kaltwasser, C. (2017). Populism : a very short introduction.
- NOS. (2011). Ministers Curaçao “niet ministeriabel” | NOS. Retrieved June 12, 2019, from



- <https://nos.nl/artikel/304861-ministers-curacao-niet-ministeriabel.html>
- NOS. (2018). Veroordeelde ex-premier Curaçao Schotte meldt zich bij gevangenis | NOS. Retrieved June 12, 2019, from <https://nos.nl/artikel/2262207-veroordeelde-ex-premier-curacao-schotte-meldt-zich-bij-gevangenis.html>
- NU.nl. (2017). Curaçaose oud-premier Gerrit Schotte krijgt drie jaar cel wegens corruptie | NU - Het laatste nieuws het eerst op NU.nl. Retrieved June 12, 2019, from <https://www.nu.nl/buitenland/4851981/curaaose-oud-premier-gerrit-schotte-krijgt-drie-jaar-cel-wegens-corruptie.html>
- Rea, A., Fischer, A., & Rose du, J. (2019). Colony. Retrieved from <https://colony.io/whitepaper.pdf>
- Rosenmöller, P., Maas, C., & Hillebrink, S. (2011). Doe Het Zelf: Rapport van de Commissie Onderzoek Curaçao, ingesteld bij Koninklijk Besluit van 8 augustus 2011. Retrieved from www.commissieonderzoekcuracao.info
- Seng, B., Hirayama, K., Katayama-Hirayama, K., Ochiai, S., & Kaneko, H. (2013). Scenario analysis of the benefit of municipal organic-waste composting over landfill, Cambodia. *Journal of Environmental Management*, 114, 216–224. <https://doi.org/10.1016/J.JENVMAN.2012.10.002>
- Sinica, V. (2012). Populism in the Caribbean: A case study of Curaçao. *International Journal of Development Issues*.
- Stiglitz, J. (2006). Joseph Stiglitz Q & A. Retrieved from http://economistsview.typepad.com/economistsview/2006/10/joseph_stiglitz.html
- Stiglitz, J. E. (2003). *The roaring nineties : a new history of the world's most prosperous decade*. W.W. Norton.
- Transparency International. (2011). Transparency International is the global civil society organisation leading the fight against corruption . Through more than 90 chapters worldwide and an international secretariat in Berlin , we raise awareness of the damaging effects of corruption and wo. Retrieved from www.transparency.org
- Transparency International. (2018). Corruption perceptions, 16. Retrieved from <http://cpi.transparency.org/cpi2013/results/>
- Vitez, O. (2019). Centralized Vs. Decentralized Organizational Structure | Chron.com. Retrieved June 12, 2019, from <https://smallbusiness.chron.com/centralized-vs-decentralized-organizational-structure-2785.html>
- Walsh, D. (2014). Bitcoin Magazine | DACs VS the Corporation. Retrieved June 12, 2019, from <https://bitcoinmagazine.com/articles/dacs-vs-the-corporation-1416342363>
- Worldpopulationreview. (2019). Curaçao Population 2019 (Demographics, Maps, Graphs). Retrieved June 11, 2019, from <http://worldpopulationreview.com/countries/curacao-population/>